

Gate (II.11), phase V2:

The interaction comparison on the space-time torus,
the renormalization dictionary, and uniform Nelson bounds

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Abstract

This document executes phase V2 of the thermal two-torus route: Statements A4 and A5 of the scoping note are proved. A single Gaussian family realizes the lattice and continuum periodic free fields on one probability space (the same-noise, different-multiplier coupling); the two interactions, written in ensemble-Wick plus finite-recombination form, are compared in L^2 with the explicit rate $O(\varepsilon_N(1 + \log(1/\varepsilon_N))^{3/2})$; principal (dispersion + missing-mode) and aliasing contributions to the squared norm are all $O(\varepsilon_N^2 \log^3(1/\varepsilon_N))$. The renormalization dictionary is proved with a rate: $\delta_N = c_{N,L} - c_{L,N}^{\text{ct}} = \frac{1}{2\pi} \log \frac{4}{\pi} + O(\varepsilon_N)$; together with the equal-point variance comparison this locks the gate's continuum polynomial to plain φ^4 , δ_∞ entering only as the offset between the two tadpoles and not as a mass counterterm (Corollary 5.2, Remark 5.3). The mean-shift interpolant of V1 is identified in closed form, $H_k(s) = \frac{\rho_k}{2} \sinh(\gamma_N(k)(s - t/2)) / \sinh(\gamma_N(k)t/2)$, whence the shift fields are bounded uniformly in N by coarse data alone and their symbols converge at the rate $2^{-r(1+\eta)/2}$ from the $D4$ envelope and an exact filter-tail estimate. The 16 August repair retains the previously omitted four-edge sector, separates folded lattice, exact-output mixed, and continuum shift symbols, takes the real part of the asymmetric mixed surrogate, and writes the mixed alias term explicitly; no unnamed remainder is used. Finally, the uniform-in- N Nelson bounds $\sup_N \|e^{-\mathcal{U}_N}\|_{L^p} < \infty$ (plain and shifted) are proved by the Part I Appendix-C pattern, with the comparison scale selected through $1 + \log(1/\varepsilon_{N_b}) \approx (b/2c_1 t)^{1/2}$ from the depth of the negative tail and the L^2 comparison of this document replacing the mollifier estimate. Every import is isolated; all constants are explicit in their dependence on (m, L, t, λ, t_0) .

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1 The coupling

Fix $L, m > 0$, $t \geq t_0 > 0$ and, after enlarging t_0 if necessary,

$$t_0 L \geq 1, \quad (\star)$$

which is what Lemma 3.1(ii) requires and is assumed throughout this document; every statement below inherits it. It is not cosmetic — see the remark following Theorem 4.3 — and it costs nothing, L being fixed and $t \geq t_0$ throughout Part II. Dual lattices:

$$D_N := \frac{2\pi}{t} \mathbb{Z} \times \Gamma_N, \quad D_\infty := \frac{2\pi}{t} \mathbb{Z} \times \Gamma_\infty, \quad \Gamma_N = \{\pi j/L : -r_N \leq j < r_N\}, \quad \Gamma_\infty = \frac{\pi}{L} \mathbb{Z}.$$

Elements are written $\kappa = (k_0, k)$, $|\kappa|^2 = k_0^2 + k^2$. Symbols:

$$\hat{c}_N(\kappa) := \frac{1}{k_0^2 + \gamma_N(k)^2} \quad (\kappa \in D_N), \quad \hat{c}_\infty(\kappa) := \frac{1}{k_0^2 + \gamma(k)^2}, \quad \hat{c}_\times(\kappa) := (\hat{c}_N \hat{c}_\infty)^{1/2}(\kappa), \quad (1)$$

with $\gamma(k)^2 = m^2 + k^2$ and $\gamma_N(k)^2 = m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2)$. Throughout, $\log_N := 1 + \log(1/\varepsilon_N)$.

Definition 1.1 (same-noise coupling). On a probability space $(\Omega, \mathbb{P})$ let $\{G(\kappa)\}_{\kappa \in D_\infty}$ be jointly Gaussian, centered, with $G(-\kappa) = \overline{G(\kappa)}$ and $\mathbb{E}[G(\kappa)\overline{G(\kappa')}] = \delta_{\kappa\kappa'}$. Define, for $z = (s, x) \in \mathbb{T}_t \times \mathbb{T}_L$ and $x \in \Lambda_N$ respectively,

$$\eta(z) := (2Lt)^{-1/2} \sum_{\kappa \in D_\infty} G(\kappa) \hat{c}_\infty(\kappa)^{1/2} e^{i\kappa \cdot z}, \quad \eta_N(z) := (2Lt)^{-1/2} \sum_{\kappa \in D_N} G_N(\kappa) \hat{c}_N(\kappa)^{1/2} e^{i\kappa \cdot z},$$

where $G_N := G$ off the *edge column* $k_e := -\pi/\varepsilon_N$, and on it

$$G_N(k_0, k_e) := 2^{-1/2} (G(k_0, k_e) + G(k_0, -k_e)), \quad k_0 \in \frac{2\pi}{t} \mathbb{Z}.$$

The two fields are understood, respectively, as the first an $L^2(\mathbb{P})$ -valued distribution (its integrals against space–time test functions and all Wick integrals below are the objects used; no pointwise continuum field is needed), the second as an honest jointly measurable random field (finitely many spatial modes; each mode has a continuous modification in s by Kolmogorov–Chentsov, since $\mathbb{E}|X_s - X_{s'}|^2 = 2(c_\omega(0) - c_\omega(s - s')) \leq |s - s'|$ by the Lipschitz bound $|c'_\omega| \leq \frac{1}{2}$, and Gaussianity upgrades this to $2n$ -th moments).

Remark (why the edge column must be symmetrized). The band is *asymmetric*: $k_e = -\pi/\varepsilon_N \in \Gamma_N$ while $-k_e = +\pi/\varepsilon_N \in \Gamma_\infty \setminus \Gamma_N$, so $D_N \neq -D_N$ and the Hermitian constraint $G(-\kappa) = \overline{G(\kappa)}$ does *not* close on D_N at that one column. Had η_N been defined with G in place of G_N it would not be real: conjugating its defining sum reindexes it over $-D_N$, every column with $|k| < \pi/\varepsilon_N$ cancels, and — using $e^{ik_e x} = e^{-ik_e x} = (-1)^{x/\varepsilon_N}$ on Λ_N — one is left with

$$\overline{\eta_N} - \eta_N = (2Lt)^{-1/2} (-1)^{x/\varepsilon_N} \sum_{k_0} \hat{c}_N(k_0, k_e)^{1/2} e^{ik_0 s} [G(k_0, -k_e) - G(k_0, k_e)],$$

a nondegenerate Gaussian: the two families are independent, so by (6) its variance equals $\frac{1}{2L} \coth(t\gamma_e/2)/\gamma_e = \varepsilon_N/(4L) + O(\varepsilon_N^3)$, where $\gamma_e := \gamma_N(k_e) = (m^2 + 4\varepsilon_N^{-2})^{1/2}$. The induced defect in the covariance would be the omission of the whole edge column, i.e. $\varepsilon_N/(8L) + O(\varepsilon_N^3)$ at coincident points — one power of ε_N *larger* than every term tracked in §4. The symmetrization of G_N is available exactly because k_e and $-k_e$ carry the same spatial function on Λ_N and the same dispersion, $\gamma_N(k_e) = \gamma_N(-k_e)$. Reality is load-bearing downstream: the pointwise bound of Lemma 7.1 is an identity between real numbers, and V4's phase estimate uses $|e^{ia} - e^{ib}| \leq |a - b|$ for real a, b .

Proposition 1.2 (laws and covariances). (i) η_N is real, and its law is that of the V1 periodic lattice ensemble $\mu_{N,t}^{\text{per}}$:

$$\mathbb{E}[\eta_N(s, x)\eta_N(s', y)] = \frac{1}{2L} \sum_{k \in \Gamma_N} e^{ik(x-y)} c_{\gamma_N(k)}(s - s'), \quad c_\omega(\tau) = \frac{\cosh(\omega(\frac{t}{2} - |\tau|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}.$$

(ii) With $C_\times(z - z') := (2Lt)^{-1} \sum_{\kappa \in D_N} \hat{c}_\times(\kappa) e^{i\kappa(z-z')}$ (and likewise C_N, C_∞ with $\hat{c}_N, \hat{c}_\infty$), the mixed covariance is

$$\mathbb{E}[\eta_N(z)\eta(z')] = C_\times(z - z') + R_N(z, z'),$$

where R_N is carried by the edge column alone,

$$R_N(z, z') = (2Lt)^{-1} \sum_{k_0} \hat{c}_\times(k_0, k_e) e^{ik_0(s-s')} \left[\frac{1}{\sqrt{2}} (e^{ik_e(x-y)} + e^{ik_e(x+y)}) - e^{ik_e(x-y)} \right],$$

and, since the bracket has modulus $\leq \sqrt{2} + 1 < 3$ and, on that column, $\hat{c}_N = (k_0^2 + \gamma_e^2)^{-1}$ while $\hat{c}_\infty = (k_0^2 + m^2 + \pi^2 \varepsilon_N^{-2})^{-1} \leq (k_0^2 + \gamma_e^2)^{-1}$ (because $\pi^2 > 4$), whence $\hat{c}_\times \leq (k_0^2 + \gamma_e^2)^{-1}$,

$$|R_N| \leq 3(2Lt)^{-1} \sum_{k_0} \frac{1}{k_0^2 + \gamma_e^2} = \frac{3}{2L} \frac{\coth(t\gamma_e/2)}{2\gamma_e} \leq \frac{3 \coth(2t_0)}{8L} \varepsilon_N$$

by (6), using $\gamma_e \geq 2\varepsilon_N^{-1}$ and $t/\varepsilon_N \geq 2t_0$ for $\varepsilon_N \leq \frac{1}{2}$, $t \geq t_0$. (iii) The edge column is best recorded not through R_N but through the equivalent symmetric description that R_N encodes. Because $\hat{c}_\infty(k_0, k_e) = \hat{c}_\infty(k_0, -k_e)$ and $e^{ik_e x} = e^{-ik_e x}$ on Λ_N , the mixed kernel is

$$\mathbb{E}[\eta_N(z)\eta(z')] = (2Lt)^{-1} \sum_{\kappa \in \tilde{D}_N} \chi(k) \hat{c}_\times(\kappa) e^{i\kappa(z-z')}, \quad \tilde{D}_N := \frac{2\pi}{t} \mathbb{Z} \times \{ \pi j/L : |j| \leq r_N \},$$

for $z \in \mathbb{T}_t \times \Lambda_N$, $z' \in \mathbb{T}_t \times \mathbb{T}_L$, with $\chi(k) = 1$ for $|k| < \pi/\varepsilon_N$, $\chi(\pm\pi/\varepsilon_N) = 2^{-1/2}$, and $\hat{c}_\times(k_0, +\pi/\varepsilon_N) := \hat{c}_\times(k_0, k_e)$: the mixed pairing runs over the symmetric band $\tilde{D}_N$, each edge momentum carrying weight $2^{-1/2}$. (In terms of the noise families, the edge column has two partners, not one: $\mathbb{E}[G_N(k_0, k_e)G(\kappa')] = 2^{-1/2}(\delta_{\kappa', (-k_0, \pi/\varepsilon_N)} + \delta_{\kappa', (-k_0, -\pi/\varepsilon_N)})$, which is what produces both the $e^{ik_e(x-y)}$ and the $e^{ik_e(x+y)}$ term of R_N .) Consequently: (a) in the second-moment sums the two edge configurations $(\pm\pi/\varepsilon_N, \mp\pi/\varepsilon_N)$ each carry $\chi^2 = \frac{1}{2}$ and together restore full weight, so those sums are unchanged; (b) In the fourth-moment cross sum, let

$$\mathcal{E}_N^{\text{edge}} := 4!(2Lt)^{-2} \left[\sum_{\substack{\kappa_i \in \tilde{D}_N^4 \\ \sum \kappa_i = 0}} \prod_i \chi(k_i) \hat{c}_\times(\kappa_i) - \sum_{\substack{\kappa_i \in D_N^4 \\ \sum \kappa_i = 0}} \prod_i \hat{c}_\times(\kappa_i) \right]. \quad (2)$$

Partition the difference by the number $|J|$ of edge momenta. Exactly three edge momenta are impossible: their sum is $\pm\pi/\varepsilon_N$ or $\pm 3\pi/\varepsilon_N$, so cancellation by a non-edge fourth momentum would require an edge momentum or a momentum outside the band. Four edge momenta are possible and must be retained.

One edge factor: it contributes $\sum_{k_0} \hat{c}_\times(k_0, k_e) \leq \frac{t\varepsilon_N}{4} \coth(2t_0)$, while the complementary three-fold sum is at most F_3 at spatial momentum π/ε_N , i.e. $\leq C(m, t_0)(tL)^2 \varepsilon_N^2 \log_N^2$ by (11); after the normalizing prefactor the total is $O(t\varepsilon_N^3 \log_N^2)$.

Two edge factors: the discrepant type is $(+\pi/\varepsilon_N, -\pi/\varepsilon_N)$. With $f(k_0) := \hat{c}_\times(k_0, k_e)$ this class is $\leq (2Lt)^{-2} \binom{4}{2} \|f*f\|_\infty \sum_w F_2(0, -w)$, and $\|f*f\|_\infty \leq \sum_{k_0} f(k_0)^2 \leq \sum_{k_0} (k_0^2 + \gamma_e^2)^{-2} = O(t\varepsilon_N^3)$, giving $O(t\varepsilon_N^3)$ after normalization.

Four edge factors: *exact spatial conservation requires two momenta $+\pi/\varepsilon_N$ and two momenta $-\pi/\varepsilon_N$. There are $\binom{4}{2} = 6$ orders and each has weight $(2^{-1/2})^4 = 1/4$. Their contribution is*

$$\Delta_{4e} := 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_{i=1}^4 f(k_{0,i}). \quad (3)$$

Put $A := \sum_{k_0} f(k_0)$ and $B := \sum_{k_0} f(k_0)^2$. Since $f(k_0) \leq (k_0^2 + \gamma_e^2)^{-1}$, $\gamma_e \geq 2\varepsilon_N^{-1}$, and $t/\varepsilon_N \geq 2t_0$,

$$A \leq \frac{t\varepsilon_N}{4} \coth(2t_0), \quad B \leq \gamma_e^{-2} A \leq \frac{t\varepsilon_N^3}{16} \coth(2t_0).$$

For convolution on $(2\pi/t)\mathbb{Z}$,

$$\sum_{\sum_i k_{0,i}=0} \prod_i f(k_{0,i}) = \|f * f\|_2^2 \leq \|f * f\|_\infty \|f * f\|_1 \leq BA^2.$$

Consequently

$$0 \leq \Delta_{4e} \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5. \quad (4)$$

Combining the three possible sectors gives

$$|\mathcal{E}_N^{\text{edge}}| \leq C(m, t_0, L) t \varepsilon_N^3 \log^2 N + \Delta_{4e}. \quad (5)$$

This replaces the false assertion that all sectors with $|J| \geq 3$ are impossible.

Proof. Reality of η_N . Off the edge column both κ and $-\kappa$ lie in D_N and $G_N(-\kappa) = \overline{G_N(\kappa)}$, $\hat{c}_N(-\kappa) = \hat{c}_N(\kappa)$, so those terms pair into a real sum. On the edge column the relevant involution of D_N is $(k_0, k_e) \mapsto (-k_0, k_e)$, because $e^{ik_e x} = e^{-ik_e x}$ for $x \in \Lambda_N$; and by the edge-column convention of Definition 1.1,

$$\begin{aligned} G_N(-k_0, k_e) &= 2^{-1/2} (G(-k_0, k_e) + G(-k_0, -k_e)) \\ &= 2^{-1/2} \overline{(G(k_0, -k_e) + G(k_0, k_e))} = \overline{G_N(k_0, k_e)}, \end{aligned}$$

while $\hat{c}_N(-k_0, k_e) = \hat{c}_N(k_0, k_e)$. Hence that column is real too, and η_N is real. (Without the symmetrization of G_N this step fails; see the remark following Definition 1.1.)

Orthonormality of G_N on D_N . $\mathbb{E}[G_N(\kappa)\overline{G_N(\kappa')}] = \delta_{\kappa\kappa'}$: off the edge this is the hypothesis, on the edge $\frac{1}{2}(\delta + 0 + 0 + \delta) = \delta_{k_0 k_0'}$, and an edge/non-edge pair is orthogonal since $\pm k_e \neq k$ for $k \in \Gamma_N \setminus \{k_e\}$. Likewise, using $\mathbb{E}[G(\kappa)G(\kappa')] = \delta_{\kappa, -\kappa'}$,

$$\mathbb{E}[G_N(k_0, k_e)G_N(k_0', k_e)] = \frac{1}{2}(0 + \delta_{k_0, -k_0'} + \delta_{k_0, -k_0'} + 0) = \delta_{k_0, -k_0'},$$

so the edge column contributes to $\mathbb{E}[\eta_N(z)\eta_N(z')]$ the term

$$(2Lt)^{-1} \sum_{k_0} \hat{c}_N(k_0, k_e) e^{ik_0(s-s')} e^{ik_e(x+y)},$$

and $e^{ik_e(x+y)} = e^{ik_e(x-y)}$ on Λ_N : exactly the $k = k_e$ term of the displayed kernel, with full weight. Off the edge the pairing $\kappa' = -\kappa$ is the usual one. The covariance is therefore as stated; for (i) it remains to evaluate the k_0 -sum:

$$\frac{1}{t} \sum_{k_0 \in \frac{2\pi}{t}\mathbb{Z}} \frac{e^{ik_0\tau}}{k_0^2 + \omega^2} = c_\omega(\tau) \quad (\omega > 0), \quad (6)$$

which holds because both sides are continuous in τ , the right side has Fourier coefficients $t^{-1} \int_0^t e^{-ik_0\tau} c_\omega(\tau) d\tau = t^{-1}(k_0^2 + \omega^2)^{-1}$ — integrate $(-\partial_\tau^2 + \omega^2)c_\omega = \delta_{\mathbb{T}_t}$ against $e^{-ik_0\tau}$, where the distributional identity on the circle follows from $c_\omega'' = \omega^2 c_\omega$ off 0 and the derivative jump $c_\omega'(0^+) - c_\omega'(0^-) = -1$ (direct differentiation) — and the left side is the absolutely convergent Fourier series of a continuous function. This matches V1, Remark 5.5, so the laws agree. $\square$

2 Wick structure and the two interactions

Equal-point variances (finite on the lattice):

$$v_N(t) := \mathbb{E}[\eta_N(z)^2] = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{\coth(t\gamma_N(k)/2)}{2\gamma_N(k)}, \quad c_{N,L} = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{1}{2\gamma_N(k)}.$$

Lemma 2.1 (recombination; the finite constants d). *For any real random variable X with variance v and any constant c ,*

$$:X^4:_c = :X^4:_v + 6(v-c):X^2:_v + 3(v-c)^2, \quad :X^2:_c = :X^2:_v + (v-c), \quad (7)$$

where $:X^j:_a$ denotes Wick ordering with constant a ($:X^2:_a = X^2 - a$, $:X^4:_a = X^4 - 6aX^2 + 3a^2$). Moreover

$$d_N(t) := v_N(t) - c_{N,L} = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{\coth(t\gamma_N(k)/2) - 1}{2\gamma_N(k)} = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{1}{\gamma_N(k)(e^{t\gamma_N(k)} - 1)}, \quad (8)$$

and, with $d_\infty(t)$ the same sum over Γ_∞ with γ and $\gamma_-(k) := \max(m, \frac{2}{\pi}|k|)$:

$$0 \leq d_N(t) \leq \bar{d} := \frac{1}{2L} \sum_{k \in \Gamma_\infty} \frac{1}{\gamma_-(k)(e^{t_0\gamma_-(k)} - 1)} < \infty \quad \text{uniformly in } N \text{ and } t \geq t_0,$$

and $|d_N(t) - d_\infty(t)| \leq C(m, L, t_0) \varepsilon_N^2$, uniformly in $t \geq t_0$.

Proof. (7) is polynomial algebra: $X^4 - 6cX^2 + 3c^2 = (X^4 - 6vX^2 + 3v^2) + 6(v-c)(X^2 - v) + 3(v-c)^2$. In (8), $\coth(y/2) - 1 = 2/(e^y - 1)$. Uniform bound: on the band, $\gamma_N(k) \geq \gamma_-(k)$ (from $\sin u \geq \frac{2}{\pi}u$ on $[0, \pi/2]$), $y \mapsto (y(e^{ty} - 1))^{-1}$ is decreasing in both y and t , and $\Gamma_N \subset \Gamma_\infty$ with positive summands, so $d_N(t) \leq \bar{d}$; $\bar{d} < \infty$ since its summands decay like $e^{-2t_0|k|/\pi}$. Convergence with rate: for fixed k , $f(y) := (y(e^{ty} - 1))^{-1}$ has

$$|f'(y)| = \frac{(e^{ty} - 1) + ty e^{ty}}{y^2(e^{ty} - 1)^2} \leq \frac{1}{y^2(e^{ty} - 1)} + \frac{t e^{ty}}{y(e^{ty} - 1)^2} \leq C(1+t) \frac{e^{-ty}}{\min(y, 1)^2} \\ (y \geq m, \quad ty \geq t_0 m),$$

and $0 \leq \gamma(k) - \gamma_N(k) \leq \varepsilon_N^2 k^4 / (24m)$ (free-rate document, Lemma 3.1, eq. (3.1)); the mean value theorem on $[\gamma_N(k), \gamma(k)] \subset [\gamma_-(k), \infty)$ gives per- k errors $\leq C(m)(1+t)e^{-t\gamma_-(k)}\varepsilon_N^2 k^4$, whose $\frac{1}{2L}$ -weighted sum is $\leq C(m, L, t_0)\varepsilon_N^2$ uniformly in $t \geq t_0$ (as $\sup_{t \geq t_0} (1+t)e^{-t\gamma_-/2} < \infty$ and $\sum_k e^{-t_0\gamma_-(k)/2} k^4 < \infty$); the missing modes $k \in \Gamma_\infty \setminus \Gamma_N = \{k \geq \pi/\varepsilon_N\} \cup \{k < -\pi/\varepsilon_N\}$ — note that $+\pi/\varepsilon_N$ is missing while $-\pi/\varepsilon_N$ is not, the band being asymmetric — contribute $\leq C e^{-2t_0/\varepsilon_N}$. $\square$

Definition 2.2 (the interactions, ensemble-Wick form). With $\text{Vol} := 2Lt$:

$$\mathcal{U}_N := \lambda \left[X_4 + 6 d_N(t) X_2 + 3 d_N(t)^2 \text{Vol} \right], \quad X_j := \varepsilon_N \sum_{x \in \Lambda_N} \int_0^t : \eta_N(s, x)^j :_{v_N} ds, \quad (9)$$

$$\mathcal{U} := \lambda \left[Y_4 + 6 d_\infty(t) Y_2 + 3 d_\infty(t)^2 \text{Vol} \right], \quad Y_j := \int_0^t \int_{\mathbb{T}_L} : \eta(s, x)^j : dx ds, \quad (10)$$

where Y_j is defined as the $L^2(\mathbb{P})$ limit of its spatial mode truncations $Y_j^{(K)}$ (Proposition 2.4). By Lemma 2.1, $\mathcal{U}_N$ equals the V1 interaction $\lambda \varepsilon_N \sum_x \int_0^t : \eta_N^4 :_{c_{N,L}} ds$ exactly; $\mathcal{U}$ is the continuum interaction completely Wick-ordered by the continuum field's own vacuum constant — the truncation convention; by Corollary 5.2 this coincides in the limit with the lattice-vacuum convention, with no counterterm between them.

Lemma 2.3 (Wick–Isserlis pairings). *If (X, Y) are jointly Gaussian, centered, with variances v_X, v_Y and covariance $c = \mathbb{E}[XY]$, then for $j, l \geq 0$,*

$$\mathbb{E}[:X^j :_{v_X} :Y^l :_{v_Y}] = \delta_{jl} j! c^j.$$

Proof. From $\mathbb{E}[e^{\alpha X + \beta Y}] = e^{(\alpha^2 v_X + 2\alpha\beta c + \beta^2 v_Y)/2}$ and $:e^{\alpha X} := e^{\alpha X - \alpha^2 v_X/2}$ we get

$$\mathbb{E}[:e^{\alpha X} :: e^{\beta Y} :] = e^{\alpha\beta c};$$

expand both sides ($:e^{\alpha X} := \sum_j \alpha^j :X^j:/j!$, absolutely convergent in L^2) and compare coefficients of $\alpha^j \beta^l$. $\square$

Proposition 2.4 (existence of Y_j ; positivity of the mode sums). *Let $Y_j^{(K)}$ be defined with η replaced by its truncation to spatial modes $|k| \leq K$ (a smooth Gaussian field), Wick-ordered by its own variance. Then, for $j \in \{2, 4\}$,*

$$\mathbb{E}[Y_j^{(K)} Y_j^{(K')}] = j! (2Lt)^{-(j-2)} \sum_{\substack{\kappa_1, \dots, \kappa_j \in D_\infty^{\leq K \wedge K'} \\ \kappa_1 + \dots + \kappa_j = 0}} \prod_{i=1}^j \hat{c}_\infty(\kappa_i) =: j! (2Lt)^{-(j-2)} S_j^{(K \wedge K')},$$

$S_j^{(K)}$ is nondecreasing in K and bounded (Lemma 3.1 (iii)), hence $Y_j^{(K)}$ is $L^2(\mathbb{P})$ -Cauchy and $Y_j := \lim Y_j^{(K)}$ exists, with $\mathbb{E}[Y_j^2] = j!(2Lt)^{-(j-2)} S_j^{(\infty)}$.

Proof. By Lemma 2.3, $\mathbb{E}[Y_j^{(K)} Y_j^{(K')}] = j! \iint C_{K \wedge K'}(z - z')^j dz dz'$, where $C_{K \wedge K'}$ is the covariance of the two truncations (Wick pairings only connect modes present in both fields, and the joint covariance of the two truncated fields is the min-truncated kernel). Expanding C^j in its absolutely convergent Fourier series (C^j carries the prefactor $(2Lt)^{-j}$) and integrating over $(\mathbb{T}_t \times \mathbb{T}_L)^2$ gives the displayed sum ($\iint_{(\mathbb{T}_t \times \mathbb{T}_L)^2} e^{i\Sigma\kappa \cdot z} e^{-i\Sigma\kappa \cdot z'} dz dz' = (2Lt)^2$ if $\Sigma\kappa = 0$, else 0, so the net power is $(2Lt)^{-(j-2)}$; absolute convergence of the interchange is Lemma 3.1 (iii)). All summands are positive, so $K \mapsto S_j^{(K)}$ is nondecreasing and bounded, hence convergent; then $\|Y^{(K)} - Y^{(K')}\|_2^2 = j!(2Lt)^{-(j-2)} (S^{(K)} + S^{(K')} - 2S^{(K \wedge K')}) \rightarrow 0$. $\square$

3 Covariance estimates

Lemma 3.1 (envelope, decay of convolutions, summability). *Set $\hat{c}_{\text{env}}(\kappa) := \frac{\pi^2/4}{|\kappa|^2 + m^2}$ on D_∞ . Then: (i) $\hat{c}_N \leq \hat{c}_{\text{env}}$ on D_N , $\hat{c}_\infty \leq \hat{c}_{\text{env}}$, and $\hat{c}_\times \leq \hat{c}_{\text{env}}$ on D_N . (ii) With $\Sigma_1 := \sum_{\kappa \in D_\infty} \hat{c}_{\text{env}}(\kappa)^2 < \infty$ and the two- and three-fold convolutions (sums over D_∞)*

$$F_2(\chi) := \sum_{\kappa} \hat{c}_{\text{env}}(\kappa) \hat{c}_{\text{env}}(\chi - \kappa), \quad F_3 := \hat{c}_{\text{env}} * F_2 :$$

$$F_2(\chi) \leq C_2 tL \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}, \quad F_3(\chi) \leq C_3 (tL)^2 \frac{(1 + \log(2 + |\chi|))^2}{m^2 + |\chi|^2}, \quad (11)$$

with C_2, C_3 depending on (m, t_0) (the isolated temporal-grid term uses $t \geq t_0$); $tL \geq 1$ is also assumed so that the isolated two-dimensional grid term is absorbed by the density term. No later statement treats these constants as functions of m alone. (iii) $S_4^{(\infty)} \leq \sum_{\kappa} \hat{c}_{\text{env}}(\kappa) F_3(\kappa) < \infty$ and $S_2^{(\infty)} \leq \Sigma_1 < \infty$; the same bounds hold for all the mixed and lattice sums below, aliased or not, since every symbol involved is dominated by $\hat{c}_{\text{env}}$ at its own argument.

Proof. (i) On the band, $\sin u \geq \frac{2}{\pi}u$ for $u \in [0, \pi/2]$ gives $4\varepsilon^{-2} \sin^2(\varepsilon k/2) \geq (4/\pi^2)k^2$, so $k_0^2 + \gamma_N^2 \geq k_0^2 + m^2 + (4/\pi^2)k^2 \geq (4/\pi^2)(|\kappa|^2 + m^2)$; the $\hat{c}_\times$ bound follows since a geometric mean of two numbers $\leq a$ is $\leq a$. (ii) $\Sigma_1 < \infty$: the grid D_∞ has cell area $(2\pi/t)(\pi/L)$ and $\int_{\mathbb{R}^2} (|\kappa|^2 + m^2)^{-2} d\kappa < \infty$; the standard sum–integral comparison (the summand is monotone along rays outside a bounded set) gives $\Sigma_1 \leq C(m) tL$. Throughout we use the two grid counts (same comparison; density $\frac{tL}{2\pi^2}$ per unit area):

$$\begin{aligned} \sum_{|\kappa| \leq R} \hat{c}_{\text{env}}(\kappa) &\leq c_0(m) tL (1 + \log(2 + R)), \\ \sum_{|\kappa| > R} \frac{1 + \log(2 + |\kappa|)}{(m^2 + |\kappa|^2)^2} &\leq c_0(m) tL \frac{1 + \log(2 + R)}{m^2 + R^2}, \end{aligned} \quad (12)$$

and the monotone majorization: for $R \geq |\chi|/2$, $\frac{1 + \log(2 + R)}{m^2 + R^2} \leq c_1(m) \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}$ (the function is decreasing beyond a bounded set, and on the bounded set both sides are comparable to constants). F_2 : split D_∞ into $\text{I} = \{|\kappa| \leq |\chi|/2\}$, $\text{II} = \{|\chi - \kappa| \leq |\chi|/2\}$, and the remainder III (on which both $|\kappa|, |\chi - \kappa| > |\chi|/2$). On I : $|\chi - \kappa| \geq |\chi|/2$, so $\hat{c}_{\text{env}}(\chi - \kappa) \leq c_1 \frac{\pi^2/4}{m^2 + |\chi|^2}$ and the first count in (12) applies to $\sum_{\text{I}} \hat{c}_{\text{env}}(\kappa)$. On II : symmetric. On III : for each κ let $w(\kappa)$ be whichever of $\kappa, \chi - \kappa$ has the *larger* modulus and $w'(\kappa)$ the other; then $|w| \geq |w'| > |\chi|/2$, and since $R \mapsto \frac{\pi^2/4}{m^2 + R^2}$ is decreasing, the product of the two factors is at most $(\frac{\pi^2/4}{m^2 + |w'|^2})^2$. At most two κ produce the same w' , so by the second count in (12) (without the logarithm), $\sum_{\text{III}} \leq 2 \sum_{|w'| > |\chi|/2} (\frac{\pi^2/4}{m^2 + |w'|^2})^2 \leq C(m) tL (m^2 + |\chi|^2)^{-1}$. Summing the three regions gives the F_2 bound. $F_3 = \hat{c}_{\text{env}} * F_2$: same three regions with $f = F_2, g = \hat{c}_{\text{env}}$. Region I ($|u| \leq |\chi|/2$, u the F_2 -argument): $g(\chi - u) \leq c_1 \frac{\pi^2/4}{m^2 + |\chi|^2}$ and $\sum_{|u| \leq |\chi|/2} F_2(u) \leq C_2 tL \sum_{|u| \leq |\chi|/2} \frac{1 + \log(2 + |u|)}{m^2 + |u|^2} \leq C(m) (tL)^2 (1 + \log(2 + |\chi|))^2$ (one logarithm from the summand, one from the radial integral). Region II ($|\chi - u| \leq |\chi|/2$, so $|u| \geq |\chi|/2$): $F_2(u) \leq C(m) tL c_1 \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}$ by monotone majorization, and $\sum_{|\chi - u| \leq |\chi|/2} \hat{c}_{\text{env}}(\chi - u) \leq c_0 tL (1 + \log(2 + |\chi|))$. Region III (both moduli $> |\chi|/2$): with w the larger-modulus argument as before, majorize the w -factor's decay profile at $|w'|$ (monotone majorization, valid since $|w| \geq |w'|$) so that the product is dominated by $C(m) tL \frac{1 + \log(2 + |w'|)}{(m^2 + |w'|^2)^2}$, and apply the second count in (12) over $\{|w'| > |\chi|/2\}$: $\text{III} \leq C(m) (tL)^2 \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}$. Summing the regions gives the F_3 bound. (iii) $S_4^{(\infty)} \leq \sum_{\kappa} \hat{c}_{\text{env}}(\kappa) F_3(-\kappa) \leq C(tL)^2 \sum_{\kappa} \frac{(1 + \log(2 + |\kappa|))^2}{(m^2 + |\kappa|^2)^2} < \infty$. $\square$

Lemma 3.2 (symbol differences). *For $\kappa \in D_N$:*

$$|\hat{c}_N(\kappa) - \hat{c}_\infty(\kappa)| \leq \frac{\varepsilon_N^2 k^4}{12} \hat{c}_N(\kappa) \hat{c}_\infty(\kappa) \leq \frac{\pi^4}{192} \varepsilon_N^2 k^4 \hat{c}_{\text{env}}(\kappa) \frac{1}{m^2 + |\kappa|^2}, \quad (13)$$

and $|\hat{c}_\times - \hat{c}_\infty| \leq |\hat{c}_N - \hat{c}_\infty|$ (from $|\sqrt{a} - \sqrt{b}| = \frac{|a-b|}{\sqrt{a} + \sqrt{b}}$ and $\sqrt{\hat{c}_\infty} \leq \sqrt{\hat{c}_N} + \sqrt{\hat{c}_\infty}$).

Proof. $\hat{c}_N - \hat{c}_\infty = (\gamma^2 - \gamma_N^2) \hat{c}_N \hat{c}_\infty$ and $0 \leq k^2 - 4\varepsilon^{-2} \sin^2(\varepsilon k/2) \leq \varepsilon^2 k^4/12$ (free-rate document, (3.5)); then insert the envelope twice. $\square$

4 The fourth-moment comparison (A4(a))

Lemma 4.1 (the three sums). *With X_4, Y_4 as in Definition 2.2,*

$$\begin{aligned}\mathbb{E}[X_4^2] &= 4! (2Lt)^{-2} \sum_{\substack{\kappa_i \in D_N^4, \Sigma k_0=0 \\ \Sigma k \equiv 0 \pmod{2\pi/\varepsilon_N}}} \prod_{i=1}^4 \hat{c}_N(\kappa_i), \\ \mathbb{E}[Y_4^2] &= 4! (2Lt)^{-2} \sum_{\kappa_i \in D_\infty^4, \Sigma \kappa=0} \prod_{i=1}^4 \hat{c}_\infty(\kappa_i).\end{aligned}\tag{14}$$

Define the asymmetric-band surrogate

$$S_{\times, N}^D := 4! (2Lt)^{-2} \sum_{\kappa_i \in D_N^4, \Sigma \kappa=0} \prod_{i=1}^4 \hat{c}_\times(\kappa_i).$$

Proposition 1.2(iii) says that the actual mixed pairing runs over the symmetric band $\tilde{D}_N = \frac{2\pi}{t}\mathbb{Z} \times \{\pi j/L : |j| \leq r_N\}$ with edge weight χ . Therefore the exact cross identity is

$$\mathbb{E}[X_4 Y_4] = 4! (2Lt)^{-2} \sum_{\kappa_i \in \tilde{D}_N^4, \Sigma \kappa=0} \prod_{i=1}^4 \chi(k_i) \hat{c}_\times(\kappa_i) = S_{\times, N}^D + \mathcal{E}_N^{\text{edge}}.$$

Thus the asymmetric sum is never asserted to be the true cross moment. The bound (5), including the four-edge term (3), is carried explicitly into the proof of Theorem 4.3. In the second-chaos analogue the two descriptions agree exactly, by Proposition 1.2(iii)(a). The lattice constraint splits into the principal part $\Sigma k = 0$ and the alias part $\Sigma k \in \{\pm 2\pi/\varepsilon_N, -4\pi/\varepsilon_N\}$ (no other values occur, since each $k_i \in [-\pi/\varepsilon_N, \pi/\varepsilon_N]$).

Proof. By Lemma 2.3, $\mathbb{E}[X_4^2] = 4! \varepsilon_N^2 \sum_{x, x'} \iint C_N(s-s', x-x')^4$, and $\varepsilon_N \sum_{u \in \Lambda_N} e^{iKu} = 2L \cdot \mathbf{1}[K \in (2\pi/\varepsilon_N)\mathbb{Z}]$, $\int_0^t e^{ik_0\tau} d\tau = t \mathbf{1}[k_0 = 0]$ applied to the absolutely convergent (Lemma 3.1) Fourier expansion of C_N^4 ; the Vol-factors combine to $(2Lt)^2 (2Lt)^{-4}$. For the cross term, $\int_{\mathbb{T}_L} dy$ enforces exact conservation of the D_∞ -labels carried by η ; by Definition 1.1 those labels run over $\tilde{D}_N$, the two edge momenta $\pm\pi/\varepsilon_N$ each carrying the weight $2^{-1/2} \hat{c}_\times$ (Proposition 1.2(iii)), and the lattice constraint $\Sigma k \equiv 0$ is then automatic. It is *not* true that modes of η outside D_N have zero covariance with η_N : the mode at $+\pi/\varepsilon_N$ does not, and that is exactly what the symmetrization of G_N creates. The alias values: $\Sigma k \in [-4\pi/\varepsilon_N, 4\pi/\varepsilon_N]$ intersected with $(2\pi/\varepsilon_N)\mathbb{Z}$. $\square$

Lemma 4.2 (alias bound).

$$\text{Al}_N := 4! (2Lt)^{-2} \sum_{\substack{\kappa_i \in D_N^4, \Sigma k_0=0 \\ \Sigma k \in \{\pm 2\pi/\varepsilon_N, -4\pi/\varepsilon_N\}}} \prod_i \hat{c}_N(\kappa_i) \leq C_{\text{al}}(m, t_0) tL \varepsilon_N^2 (1 + \log(1/\varepsilon_N))^3. \tag{15}$$

Proof. Parametrize by $(\kappa_1, \kappa_2, \kappa_3)$; then $\kappa_4 = \sigma - \kappa_1 - \kappa_2 - \kappa_3$ with $\sigma = (0, \Sigma k)$, $|\sigma| \geq 2\pi/\varepsilon_N$. Domination by the envelope and Lemma 3.1(ii):

$$\sum_{\kappa_2, \kappa_3} \hat{c}_{\text{env}}(\kappa_2) \hat{c}_{\text{env}}(\kappa_3) \hat{c}_{\text{env}}(\sigma - \kappa_1 - \kappa_2 - \kappa_3) = F_3(\sigma - \kappa_1) \leq C_3 (tL)^2 \frac{(1 + \log(2 + |\sigma - \kappa_1|))^2}{m^2 + |\sigma - \kappa_1|^2},$$

The *spatial* component of $\sigma - \kappa_1$ has modulus $\geq 2\pi/\varepsilon_N - \pi/\varepsilon_N = \pi/\varepsilon_N$ (and $\geq 3\pi/\varepsilon_N$ in the corner case), and, since $x \mapsto (1 + \log(2 + x))^2/(m^2 + x^2)$ is decreasing for $x \geq x_0(m)$,

the supremum of (11)'s right-hand side over $|\sigma - \kappa_1| \geq \pi/\varepsilon_N$ is attained at the endpoint, giving $F_3(\sigma - \kappa_1) \leq C_3(tL)^2 \varepsilon_N^2 \pi^{-2} (1 + \log(2 + 4\pi/\varepsilon_N))^2$ uniformly in κ_1 (for $\varepsilon_N \leq \varepsilon_0(m)$; the finitely many larger ε_N are absorbed into C_{al}). Next,

$$\sum_{\kappa_1 \in D_N} \hat{c}_{\text{env}}(\kappa_1) \leq \sum_{k \in \Gamma_N} \sum_{k_0 \in \frac{2\pi}{t}\mathbb{Z}} \frac{\pi^2/4}{m^2 + k_0^2 + k^2} \leq C t \sum_{k \in \Gamma_N} \frac{1}{(m^2 + k^2)^{1/2}} \leq C(m, t_0) t L \log_N$$

(one-dimensional sum–integral comparisons in k_0 , then in k up to the band radius π/ε_N). Multiply the three bounds and the prefactor $4!(2Lt)^{-2}$; there are three alias values of σ , absorbed into C_{al} . $\square$

Theorem 4.3 (A4(a): interaction L^2 comparison with rate). *Assume the standing hypothesis $tL \geq 1$ of Lemma 3.1(ii). There is $C = C(m, \lambda, t_0) < \infty$ such that for all N with $\varepsilon_N \leq \frac{1}{2}$ and all $t \geq t_0$:*

$$\|\mathcal{U}_N - \mathcal{U}\|_{L^2(\mathbb{P})} \leq C(1+t)(1+L)\varepsilon_N(1+\log(1/\varepsilon_N))^{3/2}. \quad (16)$$

Remark (the rôle of $tL \geq 1$, and where it is consumed). The hypothesis is not cosmetic. Lemma 3.1(ii) bounds $\Sigma_1 = \sum_{\kappa \in D_\infty} \hat{c}_{\text{env}}(\kappa)^2$ by $C(m) tL$ through a sum–integral comparison on a grid of density $tL/2\pi^2$ plus an isolated maximal term, and that maximal term is absorbable only when $tL \gtrsim 1$. As $t \downarrow 0$ at fixed L the k_0 -spacing $2\pi/t$ diverges, only the row $k_0 = 0$ survives, and Σ_1 saturates at a t -independent value while $tL \rightarrow 0$: numerically ($m = L = 1$) $\Sigma_1 = 7.658, 6.230, 6.202, 6.201, 6.201$ at $t = 4, 1, \frac{1}{4}, \frac{1}{20}, \frac{1}{100}$, so $\Sigma_1/(tL) = 1.9, 6.2, 24.8, 124, 620$ grows like $1/t$. Hence (16) must not be invoked at arbitrarily small t . The one place in the chain that would otherwise do so is V4, Theorem 5.2(a); it is stated there for $t \geq t_*$ with $t_* > 0$ fixed, which is all the scoping note's Laplace-transfer lemma requires (see the amended hypothesis (i) of that lemma). Throughout Part II L is fixed and $t \geq t_0$, so $tL \geq 1$ costs nothing after enlarging t_0 .

Proof. Throughout, two one-dimensional sum–integral comparisons are used (both standard: even unimodal summands, density $\frac{t}{2\pi}$ resp. $\frac{L}{\pi}$ plus the maximal term; $j \in \{0, 1, 2\}$ and any $s > 1/2$ (used below at $s = 2, 3, 4$), $a^2 := m^2 + k^2$):

$$\sum_{k_0 \in \frac{2\pi}{t}\mathbb{Z}} \frac{(1 + \log(2 + |\kappa|))^j}{(m^2 + k_0^2 + k^2)^s} \leq C_s(m, t_0) t \frac{(1 + \log(2 + |k|))^j}{a^{2s-1}}, \quad (17)$$

(split the sum at $|k_0| \leq a$, where $\log(2 + |\kappa|) \leq C(1 + \log(2 + |k|))$, and use $\int_a^\infty v^{-2s}(1 + \log(2 + v))^j dv \leq C_{j,s} a^{1-2s}(1 + \log(2 + a))^j$ on the rest — valid for all $a > 0$ and $s > 1/2$, splitting at $\max(a, 1)$; the isolated maximal term of the near part is absorbed via $1/a \leq 1/m \leq t/(mt_0)$, which is the sole source of the t_0 -dependence of C_s), and

$$\begin{aligned} \sum_{k \in \Gamma_N} \frac{k^4(1 + \log(2 + |k|))^2}{(m^2 + k^2)^{5/2}} &\leq C(m) L \log_N^3, \\ \sum_{|k| \geq \pi/\varepsilon_N} \frac{(1 + \log(2 + |k|))^2}{(m^2 + k^2)^{3/2}} &\leq C(m) L \varepsilon_N^2 \log_N^2 \end{aligned} \quad (18)$$

(the first summand decays like $|k|^{-1} \log^2 |k|$, so its band sum is $O(\log^3)$; the second is a convergent tail of $|k|^{-3} \log^2 |k|$).

Fourth chaos. By Lemma 4.1 and the decomposition of $\mathbb{E}[X_4^2]$ into principal and alias parts,

$$\begin{aligned} \|X_4 - Y_4\|_2^2 &= 4! (2Lt)^{-2} B_4 + \text{Al}_N - 2\mathcal{E}_N^{\text{edge}}, \\ B_4 &:= \sum_{\substack{\kappa \in D_N^4 \\ \Sigma \kappa = 0}} \left[\prod_i \hat{c}_N - 2 \prod_i \hat{c}_\times + \prod_i \hat{c}_\infty \right] \\ &\quad + \sum_{\substack{\kappa \in D_\infty^4 \setminus D_N^4 \\ \Sigma \kappa = 0}} \prod_i \hat{c}_\infty. \end{aligned}$$

For the first sum, telescope each product difference over the four factors:

$$\left| \prod_i \hat{c}_a(\kappa_i) - \prod_i \hat{c}_\infty(\kappa_i) \right| \leq \sum_{i=1}^4 |\hat{c}_a - \hat{c}_\infty|(\kappa_i) \prod_{j \neq i} \hat{c}_{\text{env}}(\kappa_j), \quad a \in \{N, \times\},$$

valid on D_N^4 by Lemma 3.1(i); with $|\hat{c}_\times - \hat{c}_\infty| \leq |\hat{c}_N - \hat{c}_\infty|$ (Lemma 3.2) and the i -symmetry, and dominating the constrained three-fold sums by F_3 (positive summands, larger index set),

$$\begin{aligned} |B_4| &\leq 12 T_1 + 4 T_2, \\ T_1 &:= \sum_{\kappa_1 \in D_N} |\hat{c}_N - \hat{c}_\infty|(\kappa_1) F_3(-\kappa_1), \quad T_2 := \sum_{\kappa_1 \in D_\infty \setminus D_N} \hat{c}_{\text{env}}(\kappa_1) F_3(-\kappa_1). \end{aligned}$$

By Lemma 3.2, $|\hat{c}_N - \hat{c}_\infty|(\kappa) \leq \frac{\pi^6}{768} \varepsilon_N^2 k^4 (m^2 + |\kappa|^2)^{-2}$, so with (11), (17) (at $s = 3$) and (18),

$$\begin{aligned} T_1 &\leq C(m) (tL)^2 \varepsilon_N^2 \sum_{\kappa \in D_N} \frac{k^4 (1 + \log(2 + |\kappa|))^2}{(m^2 + |\kappa|^2)^3} \\ &\leq C(m, t_0) (tL)^2 t \varepsilon_N^2 \sum_{k \in \Gamma_N} \frac{k^4 (1 + \log(2 + |k|))^2}{(m^2 + k^2)^{5/2}} \leq C (tL)^3 \varepsilon_N^2 \log_N^3, \end{aligned}$$

and with (17) at $s = 2$:

$$\begin{aligned} T_2 &\leq C(m) (tL)^2 \sum_{\substack{\kappa \in D_\infty \\ |k| \geq \pi/\varepsilon_N}} \frac{(1 + \log(2 + |\kappa|))^2}{(m^2 + |\kappa|^2)^2} \\ &\leq C(m, t_0) (tL)^2 t \sum_{|k| \geq \pi/\varepsilon_N} \frac{(1 + \log(2 + |k|))^2}{(m^2 + k^2)^{3/2}} \leq C (tL)^3 \varepsilon_N^2 \log_N^2. \end{aligned}$$

With the prefactor, the alias bound (15), and the separately tracked edge bound (5),

$$\begin{aligned} \|X_4 - Y_4\|_2^2 &\leq 4! (2Lt)^{-2} |B_4| + \text{Al}_N + 2|\mathcal{E}_N^{\text{edge}}| \\ &\leq C(m, t_0) tL \varepsilon_N^2 \log_N^3. \end{aligned} \tag{19}$$

Here $t_0 L \geq 1$, $\varepsilon_N \leq 1/2$, and $\log_N \geq 1$ imply that both $t \varepsilon_N^3 \log_N^2$ and the explicit $9t \coth(2t_0)^3 \varepsilon_N^5 / (256L^2)$ are bounded by the last line after enlarging a constant depending only on (m, t_0) .

Second chaos. Here the principal part is an exact square: by Lemma 2.3 (as in Lemma 4.1, with net prefactor $(2Lt)^0$), using $\hat{c}_\times(\kappa) \hat{c}_\times(-\kappa) = \hat{c}_N(\kappa) \hat{c}_\infty(\kappa)$ and evenness of all symbols,

$$\|X_2 - Y_2\|_2^2 = 2 \sum_{\kappa \in D_N} (\hat{c}_N - \hat{c}_\infty)^2(\kappa) + 2 \sum_{\kappa \in D_\infty \setminus D_N} \hat{c}_\infty(\kappa)^2,$$

exactly: no alias term and no edge remainder survive. Indeed, on the edge column the lattice alias pairing $k_1 = k_2 = k_e$ — forced by the unique alias value $-2\pi/\varepsilon_N$, since $-k_e \notin \Gamma_N$ — supplies precisely the principal pairing that the asymmetry withholds, so $\mathbb{E}[X_2^2] = 2 \sum_{\kappa \in D_N} \hat{c}_N(\kappa)^2$ with the alias *replacing*, not augmenting, the missing term; and by Proposition 1.2(iii)(a) the two symmetrized cross configurations $(\pm\pi/\varepsilon_N, \mp\pi/\varepsilon_N)$, each of weight $\chi^2 = \frac{1}{2}$, restore full weight, so $\mathbb{E}[X_2 Y_2] = 2 \sum_{\kappa \in D_N} \hat{c}_N \hat{c}_\infty$. With $\mathbb{E}[Y_2^2] = 2 \sum_{D_\infty} \hat{c}_\infty^2$, the display follows by subtraction. The square term is bounded by Lemma 3.2 (squared: $(\hat{c}_N - \hat{c}_\infty)^2 \leq C \varepsilon_N^4 k^8 (m^2 + |\kappa|^2)^{-4}$) together with (17) at $s = 4$, which gives $\sum_{k_0} \leq C(m, t_0) t (m^2 + k^2)^{-7/2}$; the remaining k -sum $\sum_{\Gamma_N} k^8 (m^2 + k^2)^{-7/2}$ has summand $\sim |k|$ and band value $O(L \varepsilon_N^{-2})$, so $2 \sum_{D_N} (\hat{c}_N - \hat{c}_\infty)^2 \leq C(m, t_0) t L \varepsilon_N^2$ — the same order as the missing-mode term, not subleading; the missing modes give $\leq C(m, t_0) t L \varepsilon_N^2$ by (17) ($s = 2$) followed by the log-free tail $\sum_{|k| \geq \pi/\varepsilon_N} (m^2 + k^2)^{-3/2} \leq C(m) L \varepsilon_N^2$, which is the second line of (18) with the numerator $(1 + \log(2 + |k|))^2$ deleted — the same integral comparison, run without the logarithm. Hence $\|X_2 - Y_2\|_2^2 \leq C(m, t_0) t L \varepsilon_N^2$.

Assembly. By Definition 2.2,

$$\|\mathcal{U}_N - \mathcal{U}\|_2 \leq \lambda \left[\|X_4 - Y_4\|_2 + 6\bar{d} \|X_2 - Y_2\|_2 + 6|d_N - d_\infty| \|Y_2\|_2 + 3|d_N^2 - d_\infty^2| (2Lt) \right],$$

with $\bar{d}$ from Lemma 2.1, $\|Y_2\|_2^2 = 2S_2^{(\infty)} \leq 2\Sigma_1 \leq C(m) t L$, and $|d_N - d_\infty| \leq C \varepsilon_N^2$, $|d_N^2 - d_\infty^2| \leq 2\bar{d} C \varepsilon_N^2$. All four terms are $\leq C(m, \lambda, t_0)(1+t)(1+L)\varepsilon_N \log_N^{3/2}$ (for the last, $\varepsilon_N^2 t \leq \varepsilon_N(1+t)$ since $\varepsilon_N \leq 1$), which is (16). $\square$

Remark 4.4. All three mechanisms — dispersion (T_1), missing modes (T_2), aliasing — contribute $O(\varepsilon_N^2 \log_N^{2-3})$ to the squared norm; none dominates parametrically, and no exponent was given away. For A6 only $\|\mathcal{U}_N - \mathcal{U}\|_{L^2} \rightarrow 0$ is consumed; the explicit rate feeds the scale selection in Theorem 7.2 and is recorded for any later quantitative use. No numerical input is used anywhere in this section.

5 The renormalization dictionary (A4(b))

Theorem 5.1 (the dictionary constant, with rate). *Let $c_{L,N}^{\text{ct}} := \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{1}{2\gamma(k)}$ (the Γ_N -truncated continuum vacuum constant). Then*

$$\left| \delta_N - \frac{1}{2\pi} \log \frac{4}{\pi} \right| \leq C(m, L) \varepsilon_N \quad (\varepsilon_N \leq \tfrac{1}{2}), \quad \delta_N := c_{N,L} - c_{L,N}^{\text{ct}}, \quad (20)$$

with $C(m, L)$ explicit from the proof ($C = \frac{1}{8L} + \frac{\pi C_0}{8L} + C'(m)$, $C_0 < 6$: the Riemann step contributes $C_0 \frac{\pi}{2} \Delta u$ with $\Delta u = \pi \varepsilon_N / (2L)$, carried through the prefactor $\frac{1}{2\pi}$ of Step 1, i.e. $\frac{1}{2\pi} C_0 \frac{\pi}{2} \cdot \frac{\pi \varepsilon_N}{2L} = \frac{\pi C_0}{8L} \varepsilon_N$).

Proof. Step 1: exact reduction. With $k_j = \pi j / L$, $u_j := \varepsilon_N k_j / 2 = \pi j / (2r_N)$ and $\tilde{\delta} := m \varepsilon_N / 2$: $\gamma_N(k_j) = \frac{2}{\varepsilon_N} (\tilde{\delta}^2 + \sin^2 u_j)^{1/2}$ and $\gamma(k_j) = \frac{2}{\varepsilon_N} (\tilde{\delta}^2 + u_j^2)^{1/2}$ exactly, so

$$\delta_N = \frac{1}{4L} \sum_{j=-r_N}^{r_N-1} \left(\frac{1}{\gamma_N} - \frac{1}{\gamma} \right) (k_j) = \frac{1}{2\pi} \sum_{j=1}^{r_N-1} g_{\tilde{\delta}}(u_j) \Delta u + E_{\text{edge}},$$

where $\Delta u = \pi / (2r_N)$, $g_{\tilde{\delta}}(u) := (\delta^2 + \sin^2 u)^{-1/2} - (\delta^2 + u^2)^{-1/2}$ (the $j = 0$ term vanishes; the terms $\pm j$, $1 \leq j \leq r_N - 1$, are equal and were paired, producing the factor $2 \cdot \frac{1}{4L} \cdot \frac{\varepsilon_N}{2} = \frac{\Delta u}{2\pi}$ per term). The single unpaired edge term $j = -r_N$ is $E_{\text{edge}} = \frac{1}{4L} \cdot \frac{\varepsilon_N}{2} g_{\tilde{\delta}}(\pi/2)$, and $0 \leq g_{\tilde{\delta}}(\pi/2) \leq 1$, so $0 \leq E_{\text{edge}} \leq \varepsilon_N / (8L)$.

Step 2: uniform C^1 bound. Write $g_\delta(u) = \psi(\sin^2 u) - \psi(u^2)$ with $\psi(a) = (\delta^2 + a)^{-1/2}$. Then

$$g'_\delta(u) = [\psi'(\sin^2 u) - \psi'(u^2)] \sin 2u + \psi'(u^2) (\sin 2u - 2u).$$

On $(0, \pi/2]$: $|\psi'(a) - \psi'(b)| \leq \frac{3}{4}(\min(a, b))^{-5/2}|a - b|$ (from $\psi''(a) = \frac{3}{4}(\delta^2 + a)^{-5/2}$ and the mean value theorem), $u^2 - \sin^2 u \leq u^4/3$, $\sin^2 u \geq (2u/\pi)^2$, $|\sin 2u| \leq 2u$, $|\psi'(u^2)| \leq \frac{1}{2}u^{-3}$, $|\sin 2u - 2u| \leq \frac{(2u)^3}{6}$; hence

$$|g'_\delta(u)| \leq \frac{3}{4}\left(\frac{\pi}{2u}\right)^5 \cdot \frac{u^4}{3} \cdot 2u + \frac{1}{2}u^{-3} \cdot \frac{8u^3}{6} = \frac{1}{2}\left(\frac{\pi}{2}\right)^5 + \frac{2}{3} =: C_0 < 6,$$

uniformly in $u \in (0, \pi/2]$ and $\delta \geq 0$ (and $g_\delta(0^+) = 0$). Therefore the left-endpoint Riemann sum obeys

$$\left| \sum_{j=1}^{r_N-1} g_{\tilde{\delta}}(u_j) \Delta u - \int_0^{\pi/2} g_{\tilde{\delta}}(u) du \right| \leq C_0 \frac{\pi}{2} \Delta u \leq C \varepsilon_N.$$

Step 3: removing $\tilde{\delta}$. With $\varphi(a) := a^{-1/2} - (\delta^2 + a)^{-1/2} = \int_0^{\delta^2} \frac{1}{2}(s + a)^{-3/2} ds \geq 0$: $g_0(u) - g_\delta(u) = \varphi(\sin^2 u) - \varphi(u^2) \in [0, (u^2 - \sin^2 u) \max_{[\sin^2 u, u^2]} |\varphi'|]$, and $|\varphi'(a)| = \frac{1}{2}|a^{-3/2} - (\delta^2 + a)^{-3/2}| \leq \min(\frac{1}{2}a^{-3/2}, \frac{3}{4}\delta^2 a^{-5/2})$. For $u \leq \delta$ use the first bound: $g_0 - g_\delta \leq \frac{u^4}{3} \cdot \frac{1}{2}(2u/\pi)^{-3} \leq Cu$; for $u \geq \delta$ use the second: $g_0 - g_\delta \leq \frac{u^4}{3} \cdot \frac{3}{4}\delta^2(2u/\pi)^{-5} \leq C\delta^2/u$. Hence

$$0 \leq \int_0^{\pi/2} (g_0 - g_{\tilde{\delta}}) \leq C\tilde{\delta}^2 + C\tilde{\delta}^2 \log \frac{\pi}{2\tilde{\delta}} \leq C'\varepsilon_N^2 \log(1/\varepsilon_N) \quad (\varepsilon_N \leq \frac{1}{2}).$$

Step 4: the integral. $\int_0^{\pi/2} (\csc u - u^{-1}) du = [\log \tan(u/2) - \log u]_0^{\pi/2} = -\log \frac{\pi}{2} - \log \frac{1}{2} = \log \frac{4}{\pi}$, since $\tan(u/2)/u \rightarrow \frac{1}{2}$ as $u \downarrow 0$ and $\tan(\pi/4) = 1$. Combining Steps 1–4 proves (20). (V0: the numeric value 0.0384461803 was confirmed to 3.6×10^{-10} ; the observed approach $\approx \varepsilon_N^2$ is faster than the proved $O(\varepsilon_N)$ — the proof's Riemann-sum step is first-order because g'_0 does not vanish at $\pi/2$; we do not need to sharpen it.) $\square$

Corollary 5.2 (the convention lock: no counterterm survives). *Let $:\cdot^{\text{tr}}$ denote the continuum Wick convention defined by the Γ -truncation limit (subtract $c_{L,K}^{\text{ct}}$ before $K \rightarrow \infty$). Then the continuum interaction $\mathcal{U}$ of Definition 2.2 is the truncation-convention quartic,*

$$\mathcal{U} = \lambda \int_0^t \int_{\mathbb{T}_L} :\eta^4:_{\text{tr}},$$

with no quadratic and no constant residue: the gate's continuum polynomial is plain φ^4 . Equivalently, the lattice-vacuum convention of the model sheet and the continuum truncation convention agree in the limit: although the two Wick constants differ by $\delta_\infty = \frac{1}{2\pi} \log \frac{4}{\pi} > 0$ (Theorem 5.1), the two fields' equal-point variances differ by the same δ_∞ , and the two offsets cancel.

Proof. By Definition 2.2 and Lemma 2.1 (recombination in the direction $v \rightarrow c$), at spatial truncation K ,

$$\frac{1}{\lambda} \mathcal{U}^{(K)} = Y_4^{(K)} + 6d_\infty Y_2^{(K)} + 3d_\infty^2 \text{Vol} = \int_0^t \int_{\mathbb{T}_L} :\eta^4:_{\text{tr}}, \quad c = v_\infty^{(K)} - d_\infty,$$

where $v_\infty^{(K)}(t) = \frac{1}{2L} \sum_{|k| \leq K} \coth(t\gamma(k)/2)/(2\gamma(k))$ is the truncated continuum field's own variance (the chaoses $Y_j^{(K)}$ are ordered by it, Proposition 2.4). Now

$$v_\infty^{(K)} - \frac{1}{2L} \sum_{|k| \leq K} \frac{\coth(t\gamma/2) - 1}{2\gamma} = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{2\gamma(k)} = c_{L,K}^{\text{ct}},$$

so, with $d_\infty^{(K)} := \frac{1}{2L} \sum_{|k| \leq K} [\gamma(k)(e^{t\gamma(k)} - 1)]^{-1}$ the K -truncated continuum thermal constant, $c = c_{L,K}^{\text{ct}} + (d_\infty^{(K)} - d_\infty)$ with

$$0 \leq d_\infty - d_\infty^{(K)} = \frac{1}{2L} \sum_{|k| > K} \frac{1}{\gamma(k)(e^{t\gamma(k)} - 1)} \leq C(m, L, t_0) e^{-t_0 K}$$

(using $\gamma(k) \geq |k|$), and letting $K \rightarrow \infty$ gives the display.

The second formulation is the same computation read on the lattice side. By (8), $d_N = v_N - c_{N,L}$ with v_N the lattice field's own variance, so

$$v_N - v_\infty^{(K)} = (c_{N,L} - c_{L,K}^{\text{ct}}) + (d_N - d_\infty^{(K)}) = \delta_N + (d_N - d_\infty^{(K)}) = \delta_N + O(\varepsilon_N^2) \rightarrow \delta_\infty$$

with the K -truncation taken to be Γ_N itself (so that $c_{L,K}^{\text{ct}} = c_{L,N}^{\text{ct}}$ and the displayed identity is exact), by Theorem 5.1 and Lemma 2.1. Taking instead the symmetric band $\{|k| \leq \pi/\varepsilon_N\}$ adds the single edge mode $k = +\pi/\varepsilon_N$, worth $1/(4L\gamma(\pi/\varepsilon_N)) = \varepsilon_N/(4\pi L) + O(\varepsilon_N^3)$, so that the rate becomes $\delta_N + O(\varepsilon_N)$; the limit δ_∞ , which is all the corollary consumes, is unchanged. The excess tadpole of the lattice field is thus exactly the excess Wick subtraction it receives, which is why $\mathcal{U}_N$ and $\mathcal{U}$ — each *completely* Wick-ordered by its own field's vacuum constant — are the matched pair compared in Theorem 4.3, with no counterterm between them. $\square$

Remark 5.3 (what δ_∞ is, and is not). $\delta_\infty = \frac{1}{2\pi} \log \frac{4}{\pi}$ is a genuine, correctly computed constant: it is the offset between the two *tadpoles* $c_{N,L}$ and $c_{L,N}^{\text{ct}}$ (Theorem 5.1, confirmed numerically in V0 to 3.6×10^{-10}). It is *not* a finite mass renormalization of the gate's continuum polynomial. An earlier version of this corollary asserted the residue $\varphi^4 - 6\delta_\infty \varphi^2 + 3\delta_\infty^2$; that is false, because it compares the two Wick *constants* while holding the field fixed, whereas the lattice and continuum fields have variances differing by the same δ_∞ . The error was found in the round-2 adversarial pass (independently by two reviewers, confirmed by four verifiers) and is recorded in `strategy.md`; the correction propagates to V3 §1, whose gate endpoint is $(a, b) = (0, 0)$.

6 Mean-shift coefficients (A4(c))

Recall from V1 (Lemma 5.2, `lem:interp`): for the Weyl symbol $\xi = \varepsilon_M q + ip$ at coarse scale M , the mean-shift representation at scale $N = M + r$ shifts the interaction by the harmonic jump interpolant, per mode $H_k(s) = A_k \cosh(\gamma_N(k)s) + B_k \sinh(\gamma_N(k)s)$ with $A_k = -\rho_k/2$, $B_k = \frac{\rho_k}{2} \coth(t\gamma_N(k)/2)$; the refined momentum symbol is $\widehat{p}_N(k) = 2^{r/2} P_r(\varepsilon_M k) \widehat{p}_M(k)$ on Γ_N (coarse hat $2\pi/\varepsilon_M$ -periodic; free-rate document §1), with the $D4$ envelope $|P_r(\ell)|^2 \leq C_D(1 + |\ell|)^{-2-\eta}$, $\eta = 2 - \log_2 3$, uniformly in $r \leq \infty$ (free-rate Lemma 2.1, eq. (2.4)).

Lemma 6.1 (closed form of the interpolant). *For every $\omega > 0$ and jump ρ , the interpolant of V1, Lemma 5.2 equals*

$$H(s) = \frac{\rho}{2} \frac{\sinh(\omega(s - \frac{t}{2}))}{\sinh(\omega t/2)}, \quad s \in [0, t], \quad (21)$$

and consequently $|H(s)| \leq |\rho|/2$ for all $s \in [0, t]$, and $|\partial_\omega H(s)| \leq \frac{|\rho|}{2} t \coth(\omega t/2)$.

Proof. The hyperbolic addition formula gives

$$\sinh(\omega s - \frac{\omega t}{2}) = \sinh(\omega s) \cosh(\frac{\omega t}{2}) - \cosh(\omega s) \sinh(\frac{\omega t}{2});$$

dividing by $\sinh(\omega t/2)$ and multiplying by $\rho/2$ yields exactly

$$-\frac{\rho}{2} \cosh(\omega s) + \frac{\rho}{2} \coth\left(\frac{\omega t}{2}\right) \sinh(\omega s) = A \cosh(\omega s) + B \sinh(\omega s).$$

The bound $|H| \leq |\rho|/2$ holds because $|\sinh|$ is even and increasing and $|s - t/2| \leq t/2$. For the ω -derivative, with $u := s - t/2$,

$$\partial_\omega H(s) = \frac{\rho}{2} \frac{u \cosh(\omega u) \sinh(\frac{\omega t}{2}) - \frac{t}{2} \cosh(\frac{\omega t}{2}) \sinh(\omega u)}{\sinh^2(\omega t/2)},$$

and each of the two terms is bounded in modulus by $\frac{|\rho|}{2} \cdot \frac{t}{2} \coth(\omega t/2)$ (use $|u| \leq t/2$, $\cosh(\omega u) \leq \cosh(\omega t/2)$, $|\sinh(\omega u)| \leq \sinh(\omega t/2)$). $\square$

Definition 6.2 (shift fields). In the original field coordinates (lattice transform $\widehat{f}(k) = \varepsilon_N^{1/2} \sum_x f(x) e^{-ikx}$, inverse $f(x) = \frac{\varepsilon_N^{1/2}}{2L} \sum_k \widehat{f}(k) e^{ikx}$), set $\phi_\gamma(s) := \sinh(\gamma(s - t/2)) / \sinh(\gamma t/2)$ and, using $\varepsilon_N^{1/2} 2^{r/2} = \varepsilon_M^{1/2}$,

$$\begin{aligned} h^{(N)}(s, x) &:= \frac{1}{2L} \sum_{k \in \Gamma_N} a_k^{(N)}(s) e^{ikx}, & a_k^{(N)}(s) &:= \frac{\varepsilon_M^{1/2}}{2} P_r(\varepsilon_M k) \widehat{p}_M(k) \phi_{\gamma_N(k)}(s), \\ h^{(\infty)}(s, x) &:= \frac{1}{2L} \sum_{k \in \Gamma_\infty} a_k^{(\infty)}(s) e^{ikx}, & a_k^{(\infty)}(s) &:= \frac{\varepsilon_M^{1/2}}{2} P_\infty(\varepsilon_M k) \widehat{p}_M(k) \phi_{\gamma(k)}(s). \end{aligned}$$

By Lemma 6.1, $h^{(N)}$ is precisely V1's interpolant path read in the original coordinates (same per-mode harmonic profile, boundary data $h^{(N)}(0) = -p_N/2$, $h^{(N)}(t) = +p_N/2$, periodic derivative; V1's mode relation is linear and inverts, as noted there); $h^{(\infty)}$ is its continuum counterpart with the free-rate multiplier P_∞ .

Remark (the $D4$ filter annihilates the edge column; reality of $h^{(N)}$). The mask satisfies $m_0(\pi) = 0$: by the model sheet (2.3), $m_0(\vartheta) = \left(\frac{1+e^{-i\vartheta}}{2}\right)^2 D(\vartheta)$, and $1 + e^{-i\pi} = 0$. Since $\varepsilon_M k_e = -\pi \varepsilon_M / \varepsilon_N = -\pi 2^r$, the product $P_r(\varepsilon_M k_e) = \prod_{j=1}^r m_0(2^{-j} \varepsilon_M k_e)$ contains the factor $m_0(-\pi) = 0$, so

$$P_r(\pm \pi 2^r) = 0 \quad (r \geq 1),$$

and therefore $a_{k_e}^{(N)} \equiv 0$ for every $r \geq 1$: *the coarse data never excite the edge column*. (For $r = 0$, $P_0 = 1$ and $\widehat{p}_M(-\pi/\varepsilon_M)$ is real because p_M is.) Two consequences are used below. (a) $h^{(N)}$ and $h^{(\infty)}$ are real. On Γ_∞ , and on $\Gamma_N \setminus \{k_e\}$, the Hermitian pairing $a_{-k} = \overline{a_k}$ closes (p_M real, P_r real-masked, γ_N and γ even), exactly as for η ; the one unpaired mode k_e carries the real function $e^{ik_e x} = (-1)^{x/\varepsilon_N}$ on Λ_N and the coefficient $a_{k_e}^{(N)}$, which vanishes for $r \geq 1$ and is real for $r = 0$. So no analogue of the defect of Definition 1.1 arises here. This is what licenses the shift clause of Lemma 7.1, whose pointwise identity is between *real* numbers. (b) *Where the edge remainder R_N drops out, and where it does not.* R_N is supported on the edge column, so it is annihilated by pairing against a coefficient with no edge mode — but *only where the mixed kernel enters linearly*. That is the case for V4's phase functional $\Phi_N(J_q)$, which is paired directly against $a^{(N)}$ and therefore loses its edge contribution for $r \geq 1$. It is *not* the case in Proposition 6.5, where the kernel is raised to the chaos order j . Write $R_N(z, z') = \sigma(x) \Psi(s - s', y)$ with $\sigma(x) = e^{ik_e x} = (-1)^{x/\varepsilon_N}$ on Λ_N , so that $\sigma^2 \equiv 1$. Then, e.g. for $j = 3$,

$$(C_\times + R_N)^3 - C_\times^3 = 3C_\times^2 \sigma \Psi + 3C_\times \Psi^2 + \sigma \Psi^3,$$

whose middle term carries *no* σ and is supported on the whole three-fold sumset: it is not edge-supported, and no vanishing of $a_{k_e}^{(N)}$ annihilates it. Only the kernel power 1 is edge-supported, and there $g_N = (h^{(N)})^3$, whose edge Fourier coefficient is a *convolution*, $\sum_{k_1+k_2+k_3 \equiv k_e} \prod_i a_{k_i}^{(N)}$, taken over configurations with no factor on the edge column and hence generically nonzero. In all three chaoses of Proposition 6.5 the remainder is therefore *bounded, not eliminated*; the bound is in that proposition's proof and covers $j \in \{1, 2, 3\}$ uniformly.

Lemma 6.3 (shift-field bounds; symbol convergence). *With $\bar{P}_M := \max_{k \in \Gamma_M} |\widehat{p_M}(k)|$, set*

$$\Sigma_M(p) := \frac{C_D^{1/2} \varepsilon_M^{1/2}}{4L} \left(1 + \frac{4L}{\pi \eta \varepsilon_M}\right) \bar{P}_M.$$

Then: (i) $\sup_N \sup_{s \in [0, t], x} |h^{(N)}(s, x)| \leq \Sigma_M(p)$ and $\sup_{s, x} |h^{(\infty)}(s, x)| \leq \Sigma_M(p)$; (ii) for every $j \geq 1$, $\varepsilon_N \sum_x \int_0^t |h^{(N)}|^j ds \leq 2Lt \Sigma_M(p)^j$, and likewise for $h^{(\infty)}$ with $\int_{\mathbb{T}_L} dx$; (iii) with $\theta_H := (1 + \eta)/2$,

$$\sup_{s \in [0, t]} \left[\frac{1}{2L} \sum_{k \in \Gamma_N} |a_k^{(N)}(s) - a_k^{(\infty)}(s)|^2 + \frac{1}{2L} \sum_{|k| \geq \pi/\varepsilon_N} |a_k^{(\infty)}(s)|^2 \right] \leq C_H 2^{-\theta_H r},$$

where $C_H = C(m, \eta) (1 + t \coth(t_0 m/2))^2 \varepsilon_M^{-4} (1 + \varepsilon_M)^5 \bar{P}_M^2 / L \cdot (1 + \frac{L}{\varepsilon_M})$ is independent of N and s .

Proof. (i) By Lemma 6.1, $|\phi_\gamma| \leq 1$, so

$$\begin{aligned} |h^{(N)}(s, x)| &\leq \frac{\varepsilon_M^{1/2}}{4L} \bar{P}_M \sum_{k \in \Gamma_N} |P_r(\varepsilon_M k)| \\ &\leq \frac{\varepsilon_M^{1/2}}{4L} \bar{P}_M C_D^{1/2} \left(1 + \frac{2L}{\pi} \int_0^\infty (1 + \varepsilon_M u)^{-1-\eta/2} du\right) = \Sigma_M(p), \end{aligned}$$

using the k -spacing π/L and $\int_0^\infty (1 + \varepsilon_M u)^{-1-\eta/2} du = \frac{2}{\eta \varepsilon_M}$; the same count over Γ_∞ covers $h^{(\infty)}$. (ii) is immediate from (i) ($\varepsilon_N \cdot 2r_N \cdot t = 2Lt$).

(iii) Write $\ell := \varepsilon_M k$ and split

$$a_k^{(N)} - a_k^{(\infty)} = \frac{\varepsilon_M^{1/2}}{2} \widehat{p_M}(k) \left[(P_r - P_\infty)(\ell) \phi_{\gamma_N(k)} + P_\infty(\ell) (\phi_{\gamma_N(k)} - \phi_{\gamma(k)}) \right].$$

Filter tail. $P_\infty(\ell) = P_r(\ell) \prod_{j>r} m_0(2^{-j}\ell)$, $|m_0| \leq 1$ (QMF), and $|1 - \prod_j z_j| \leq \sum_j |1 - z_j|$ for $|z_j| \leq 1$ (telescoping), while $|1 - m_0(x)| = |\int_0^x m'_0| \leq L_{m_0} |x|$ with $L_{m_0} := \sup |m'_0| \leq \frac{1}{\sqrt{2}} \sum_n n |h_n| = \frac{3+\sqrt{3}}{4}$ for the $D4$ mask. The first two members of (22) below hold for all ℓ ; the last uses the envelope of Lemma 2.1 of the free-rate document, which is proved only on the band, so (22) is asserted for $|\ell| \leq \pi 2^r$ — which is the only range in which it is used here and in V4. (It genuinely fails outside: P_r is $2^{r+1}\pi$ -periodic with $P_r(2^{r+1}\pi) = 1$, while $m_0(\pi) = 0$ forces $P_\infty(2^{r+1}\pi) = 0$, so the left side equals 1 there whereas the right side tends to 0.) Hence, for $|\ell| \leq \pi 2^r$,

$$|P_r - P_\infty|(\ell) \leq |P_r(\ell)| L_{m_0} |\ell| \sum_{j>r} 2^{-j} = L_{m_0} 2^{-r} |\ell| |P_r(\ell)| \leq C_D^{1/2} L_{m_0} 2^{-r} |\ell| (1 + |\ell|)^{-1-\eta/2}. \quad (22)$$

Its contribution to the mode sum is, with $|\phi| \leq 1$ and the band restriction $|\ell| \leq \pi 2^r$,

$$\frac{\varepsilon_M \bar{P}_M^2}{8L} C_D L_{m_0}^2 4^{-r} \sum_{k \in \Gamma_N} \ell^2 (1 + \ell)^{-2-\eta} \leq C(\eta) C_D L_{m_0}^2 \frac{\bar{P}_M^2}{4\pi} 4^{-r} (\pi 2^r)^{1-\eta} \leq C \bar{P}_M^2 2^{-r(1+\eta)},$$

by the k -count

$$\sum_{k \in \Gamma_N, \varepsilon_M |k| \leq X} G(\varepsilon_M |k|) \leq \frac{2L}{\pi \varepsilon_M} \int_0^X G + 2 \sup_{[0, X]} G$$

valid for any $G \geq 0$ that is *unimodal* on $[0, X]$ (nondecreasing then nonincreasing) — which both integrands used here are — and applied to $G(\ell) = \ell^2 (1 + \ell)^{-2-\eta}$ with $X = \pi 2^r$ ($\int_0^X \ell^{-\eta} d\ell = X^{1-\eta}/(1-\eta)$). The correction term must be $\sup_{[0, X]} G$, not $G(0)$: the left-endpoint comparison $\sum_j G(j\Delta) \leq \Delta^{-1} \int G + G(0)$ is valid only for *nonincreasing* G , whereas the integrands here vanish at 0 and increase over most of the range. The correction is harmless: it contributes $C \bar{P}_M^2 4^{-r} \sup G \leq C \bar{P}_M^2 4^{-r}$, and $4^{-r} < 2^{-r(1+\eta)}$.

Dispersion. By Lemma 6.1 and the mean value theorem in $\omega \in [\gamma_N(k), \gamma(k)] \subset [m, \infty)$, together with free-rate Lemma 3.1 (eq. (3.1)),

$$|\phi_{\gamma_N(k)}(s) - \phi_{\gamma(k)}(s)| \leq t \coth(t_0 m/2) \frac{\varepsilon_N^2 k^4}{24m} \quad \text{and always} \quad |\phi_{\gamma_N} - \phi_{\gamma}| \leq 2.$$

Split at $\ell_r := 2^{r/2}$. For $\ell \leq \ell_r$ use the first bound, noting $\varepsilon_N^2 k^4 = 4^{-r} \varepsilon_M^2 k^4 = 4^{-r} \ell^4 \varepsilon_M^{-2}$:

$$\begin{aligned} \frac{\varepsilon_M \bar{P}_M^2}{8L} C_D \left(\frac{t \coth(\frac{t_0 m}{2})}{24m \varepsilon_M^2} \right)^2 16^{-r} \sum_{k \in \Gamma_N: \ell \leq \ell_r} \ell^8 (1 + \ell)^{-2-\eta} \\ \leq C(m, \eta) \frac{(1+t)^2 \bar{P}_M^2}{\varepsilon_M^4} \left(1 + \frac{L}{\varepsilon_M} \right) \frac{1}{L} 16^{-r} 2^{\frac{r}{2}(7-\eta)}, \end{aligned}$$

by the k -count with $\int_0^X \ell^{6-\eta} d\ell = X^{7-\eta}/(7-\eta)$; the exponent is $-4r + \frac{r}{2}(7-\eta) = -\frac{r}{2}(1+\eta)$, i.e. this term is $O(2^{-r(1+\eta)/2})$. For $\ell > \ell_r$ use $|\Delta\phi| \leq 2$ and the envelope: $\frac{\varepsilon_M \bar{P}_M^2}{2L} C_D \sum_{\ell > \ell_r} (1 + \ell)^{-2-\eta} \leq C(\eta) \bar{P}_M^2 2^{-r(1+\eta)/2}$ ($\int_X^\infty (1 + \ell)^{-2-\eta} \leq X^{-1-\eta}$, and $2^{-\frac{r}{2}(1+\eta)}$ majorizes it).

Missing modes. $|k| \geq \pi/\varepsilon_N$ means $\ell > \pi 2^r$: $\frac{\varepsilon_M \bar{P}_M^2}{8L} C_D \sum_{\ell > \pi 2^r} (1 + \ell)^{-2-\eta} \leq C \bar{P}_M^2 2^{-r(1+\eta)}$.

Collecting the three contributions (each uniform in s) and absorbing the grid-count prefactors into C_H proves (iii). $\square$

Lemma 6.4 (folded, exact-output, and high-output power symbols). *Put $B_N := \Gamma_N$, $\mathfrak{R}_N := \pi/\varepsilon_N$, and let $[q]_N$ denote the unique representative of q modulo $2\mathfrak{R}_N$ in the half-open band $B_N = [-\mathfrak{R}_N, \mathfrak{R}_N) \cap (\pi/L)\mathbb{Z}$. For $j \in \{1, 2, 3\}$ define*

$$K_{N,j}^{\text{fold}}(p_0, p) := (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum_i k_{0,i} = p_0, [\sum_i k_i]_N = p}} \prod_i \hat{c}_N(\kappa_i), \quad (23)$$

$$K_{\times,j}(q_0, q) := (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum_i \kappa_i = (q_0, q)}} \prod_i \hat{c}_{\times}(\kappa_i), \quad (24)$$

$$K_{\infty,j}(q_0, q) := (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_{\infty} \\ \sum_i \kappa_i = (q_0, q)}} \prod_i \hat{c}_{\infty}(\kappa_i). \quad (25)$$

An empty sum is zero. Thus (23) is a multiplier on the lattice fundamental band, whereas (24)–(25) retain the exact continuum output. For $\varepsilon_N \leq 1/2$ and $t \geq t_0$:

$$\sup K_{N,j}^{\text{fold}} + \sup K_{\times,j} + \sup K_{\infty,j} \leq B_j(m, t_0), \quad (26)$$

$$\sup_{p_0, p \in B_N} |K_{N,j}^{\text{fold}}(p_0, p) - K_{\infty,j}(p_0, p)| \leq B'_j(m, t_0) \varepsilon_N^2 \log_N^2, \quad (27)$$

$$\sup_{p_0, p \in B_N} |K_{\times,j}(p_0, p) - K_{\infty,j}(p_0, p)| \leq B'_j(m, t_0) \varepsilon_N^2 \log_N^2, \quad (28)$$

$$\sup_{q_0, |q| \geq \mathfrak{R}_N} [K_{\times,j}(q_0, q) + K_{\infty,j}(q_0, q)] \leq B''_j(m, t_0) \varepsilon_N^2 \log_N^{j-1}. \quad (29)$$

All suprema are over the displayed temporal grid and spatial domains.

Proof. Let $\Phi_0 := \mathbf{1}_{\{0\}}$, $\Phi_1 := \hat{c}_{\text{env}}$, and $\Phi_2 := F_2$. For $j = 1$, (26) follows from $\hat{c}_{\text{env}} \leq \pi^2/(4m^2)$. For $j = 2, 3$, every exact-output sum is bounded by F_j (where F_2, F_3 are those of Lemma 3.1). The folded sum is the sum of the exact outputs $(p_0, p + 2n\mathfrak{R}_N)$ that lie in the j -fold band; there are at most $j + 1$ such integers n . Equation (11) and the normalization $(2Lt)^{-(j-1)}$ therefore prove (26).

If $|q| \geq \mathfrak{R}_N$, then for $j = 2, 3$

$$K_{\times,j}(q_0, q) + K_{\infty,j}(q_0, q) \leq 2(2Lt)^{-(j-1)} F_j(q_0, q) \leq C(m, t_0) \varepsilon_N^2 \log_N^{j-1}$$

by (11); for $j = 1$, each nonzero summand is at most $C(m^2 + q^2)^{-1} \leq C\varepsilon_N^2$. This proves (29).

It remains to work on the fundamental band. First restrict $K_{\infty,j}$ to tuples in D_N . Telescoping the j factors and using $|\hat{c}_\times - \hat{c}_\infty| \leq |\hat{c}_N - \hat{c}_\infty|$ gives, for $a = N$ or $a = \times$,

$$\left| K_{a,j}^{\text{exact}}(p) - K_{\infty,j}^{D_N}(p) \right| \leq j(2Lt)^{-(j-1)} \sum_{u \in D_N} |\hat{c}_N - \hat{c}_\infty|(u) \Phi_{j-1}(p - u). \quad (30)$$

Here $K_{N,j}^{\text{exact}}$ denotes the $n = 0$ part of (23). Lemma 3.2 yields the flat bound

$$|\hat{c}_N - \hat{c}_\infty|(u) \leq \frac{\pi^6}{768} \varepsilon_N^2 \frac{u_{\text{sp}}^4}{(m^2 + |u|^2)^2} \leq \frac{\pi^6}{768} \varepsilon_N^2. \quad (31)$$

For $j = 1$, (30) is therefore $O(\varepsilon_N^2)$. For $j = 2$, summing Φ_1 over the doubled band gives $C(m, t_0) t L \log_N$; for $j = 3$, summing Φ_2 gives $C(m, t_0) (tL)^2 \log_N^2$. After the normalization in (30),

$$\sup_{p_0, p \in B_N} |K_{a,j}^{\text{exact}}(p) - K_{\infty,j}^{D_N}(p)| \leq C(m, t_0) \varepsilon_N^2 \log_N^{j-1}. \quad (32)$$

For completeness, the missing-input estimate used next is

$$(2Lt)^{-(j-1)} \sum_{\substack{u \in D_\infty \\ |u_{\text{sp}}| \geq \mathfrak{R}_N}} \hat{c}_{\text{env}}(u) \Phi_{j-1}(p - u) \leq C(m, t_0) \varepsilon_N^2 \log_N^{j-1}, \text{quad } p_{\text{sp}} \in B_N. \quad (33)$$

For $j = 1$ this is interpreted as the single possible input and follows from $\hat{c}_{\text{env}}(p) \leq C\varepsilon_N^2$. For $j = 2, 3$, sum the temporal variable first. With $a = (m^2 + u_{\text{sp}}^2)^{1/2}$ and $b = (m^2 + (p_{\text{sp}} - u_{\text{sp}})^2)^{1/2}$, the elementary bound

$$\sum_{u_0 \in (2\pi/t)\mathbb{Z}} \frac{1}{u_0^2 + a^2} \frac{1}{(p_0 - u_0)^2 + b^2} \leq \frac{1}{a^2} \sum_{u_0} \frac{1}{(p_0 - u_0)^2 + b^2} \leq C(m, t_0) \frac{t}{a^2 b}$$

is uniform in p_0 . The spatial sum outside the band is split into $\mathfrak{R}_N \leq |u_{\text{sp}}| \leq 2\mathfrak{R}_N$ and $|u_{\text{sp}}| > 2\mathfrak{R}_N$. On the first part it is bounded by

$$Ct\mathfrak{R}_N^{-2} \sum_{0 \leq n \leq CL\mathfrak{R}_N} \frac{1}{(m^2 + (n\pi/L)^2)^{1/2}} \leq C(m, t_0)tL\mathfrak{R}_N^{-2} \log N;$$

on the second, $|p_{\text{sp}} - u_{\text{sp}}| \geq |u_{\text{sp}}|/2$ and the summand is $O(t|u_{\text{sp}}|^{-3})$, whose tail is $O(tL\mathfrak{R}_N^{-2})$. This proves (33) for $j = 2$. For $j = 3$ no appeal to a “same split” is needed. First sum F_2 over its temporal output. Since

$$\sum_{k_0 \in (2\pi/t)\mathbb{Z}} \hat{c}_{\text{env}}(k_0, k) \leq C(m, t_0) \frac{t}{(m^2 + k^2)^{1/2}},$$

Tonelli and the definition of F_2 give, with $d \in \Gamma_\infty$,

$$\sum_{q_0} F_2(q_0, d) = \sum_{k \in \Gamma_\infty} \left(\sum_{k_0} \hat{c}_{\text{env}}(k_0, k) \right) \left(\sum_{\ell_0} \hat{c}_{\text{env}}(\ell_0, d - k) \right) \quad (34)$$

$$\leq C(m, t_0)t^2 \sum_{k \in \Gamma_\infty} \frac{1}{(m^2 + k^2)^{1/2}(m^2 + (d - k)^2)^{1/2}} \quad (35)$$

$$\leq C(m, t_0)t^2 L \frac{1 + \log(2 + |d|/m)}{(m^2 + d^2)^{1/2}}. \quad (36)$$

For the last line split the k -sum into $|k| \leq |d|/2$, $|d - k| \leq |d|/2$, and their complement. On the first region the second denominator is at least $\frac{1}{2}(m^2 + d^2)^{1/2}$ and $\sum_{|k| \leq |d|/2} (m^2 + k^2)^{-1/2} \leq C(m)L[1 + \log(2 + |d|/m)]$; the second region is identical after $k \mapsto d - k$; on the complement both $|k|$ and $|d - k|$ are at least $|d|/2$, and splitting further at $|k| = 2|d|$ leaves a bounded middle part plus a tail $\sum_{|k| > 2|d|} k^{-2} \leq CL/(m + |d|)$. This proves (36).

Now use $\hat{c}_{\text{env}}(u_0, u_{\text{sp}}) \leq C(m^2 + u_{\text{sp}}^2)^{-1}$ and put $d = p_{\text{sp}} - u_{\text{sp}}$. Translation of the temporal grid and (36) imply that the left side of (33) for $j = 3$ is at most

$$\frac{C}{L} \sum_{|u_{\text{sp}}| \geq \mathfrak{R}_N} \frac{1 + \log(2 + |p_{\text{sp}} - u_{\text{sp}}|/m)}{(m^2 + u_{\text{sp}}^2)(m^2 + (p_{\text{sp}} - u_{\text{sp}})^2)^{1/2}}.$$

On $\mathfrak{R}_N \leq |u_{\text{sp}}| \leq 2\mathfrak{R}_N$, the first denominator is at least $C\mathfrak{R}_N^2$ and the remaining spatial sum is at most $CL[1 + \log(2 + \mathfrak{R}_N/m)]^2$. On $|u_{\text{sp}}| > 2\mathfrak{R}_N$, $|p_{\text{sp}} - u_{\text{sp}}| \geq |u_{\text{sp}}|/2$, so the summand is at most $C[1 + \log(2 + |u_{\text{sp}}|/m)]|u_{\text{sp}}|^{-3}$ and its tail is $CL\mathfrak{R}_N^{-2}[1 + \log(2 + \mathfrak{R}_N/m)]$. Since $\mathfrak{R}_N^{-2} = \pi^{-2}\varepsilon_N^2$, this is precisely $C(m, t_0)\varepsilon_N^2 \log_N^2$. Hence (33) holds for $j = 3$.

There are at most j choices for the missing input, so (33) bounds $K_{\infty, j} - K_{\infty, j}^{D_N}$. Finally, every nonzero folded output in (23) equals $(p_0, p + 2n\mathfrak{R}_N)$ and has spatial modulus at least $\mathfrak{R}_N$; its lattice product is envelope-dominated, so the proof of (29) bounds the finite sum of all such aliases. Adding this alias estimate and (33) to (32) proves (27)–(28). $\square$

Proposition 6.5 (shifted-interaction comparison). *Let $\Delta\mathcal{U}_N := \mathcal{U}_N(\cdot + h^{(N)}) - \mathcal{U}_N(\cdot)$ denote the V1 Wick-binomial shift of the interaction,*

$$\Delta\mathcal{U}_N = \lambda \varepsilon_N \sum_x \int_0^t \left[4h^{(N)} : \eta_N^3 :_c + 6(h^{(N)})^2 : \eta_N^2 :_c + 4(h^{(N)})^3 \eta_N + (h^{(N)})^4 \right] ds, \quad c = c_{N, L},$$

and $\Delta\mathcal{U}$ its continuum analogue with $h^{(\infty)}$ and the matched convention (each chaos existing by the truncation argument of Proposition 2.4 with deterministic bounded coefficients). Then, for $N = M + r$, $\varepsilon_N \leq \frac{1}{2}$,

$$\|\Delta\mathcal{U}_N - \Delta\mathcal{U}\|_{L^2(\mathbb{P})} \leq C \left[2^{-r(1+\eta)/4} + \varepsilon_N (1 + \log(1/\varepsilon_N)) \right],$$

with $C = C(m, \lambda, t_0, L, \varepsilon_M, \bar{P}_M) (1+t)^{5/2}$.

Proof. Write $h_N = h^{(N)}$, $h_\infty = h^{(\infty)}$ and let $I_N(F) := \varepsilon_N \sum_x \int_0^t F(s, x) ds$, $I_\infty(F) := \iint F(s, x) dx ds$. The two Wick conversions needed here are exact polynomial identities,

$$:x^3:_c = :x^3:_v + 3(v-c)x, \quad qquad :x^2:_c = :x^2:_v + (v-c).$$

Consequently, with $d_N = v_N - c_{N,L}$ and its continuum analogue,

$$G_{N,3} = 4h_N, \quad G_{N,2} = 6h_N^2, \quad G_{N,1} = 4h_N^3 + 12d_N h_N, \quad G_{N,0} = h_N^4 + 6d_N h_N^2,$$

and the same formulas with N replaced by ∞ , satisfy the exact identities

$$\begin{aligned} \Delta \mathcal{U}_N &= \lambda \left[I_N(G_{N,0}) + \sum_{j=1}^3 I_N(G_{N,j} : \eta_N^j : v_N) \right], \\ \Delta \mathcal{U} &= \lambda \left[I_\infty(G_{\infty,0}) + \sum_{j=1}^3 I_\infty(G_{\infty,j} : \eta^j : v_\infty) \right]. \end{aligned} \tag{37}$$

Thus no Wick-recombination remainder is left implicit.

Fix $j \in \{1, 2, 3\}$ and abbreviate the two stochastic integrals in (37) by $T_{N,j}$ and $T_{\infty,j}$. Isserlis gives

$$\frac{1}{j!} \mathbb{E}[(T_{N,j} - T_{\infty,j})^2] = Q_{N,j} - 2Q_{\times,j}^{\text{true}} + Q_{\infty,j}. \tag{38}$$

The true mixed kernel is $(C_\times + R_N)^j$. The asymmetric-band kernel $C_\times$ need not be real on the edge column, so define the real surrogate and the physical correction by

$$Q_{\times,j} := \Re \langle G_{N,j}, \mathbf{C}_\times^j G_{\infty,j} \rangle, \quad \mathcal{P}_{N,j} := Q_{\times,j}^{\text{true}} - Q_{\times,j} = \Re \langle G_{N,j}, [(\mathbf{C}_\times + \mathbf{R}_N)^j - \mathbf{C}_\times^j] G_{\infty,j} \rangle.$$

The last equality uses that the actual covariance pairing $Q_{\times,j}^{\text{true}}$ is real. Taking the real part can only decrease the absolute value, so the kernel sup-norm estimate below applies without loss. By Proposition 1.2(ii), $\|R_N\|_\infty \leq \varrho_N := 3 \coth(2t_0) \varepsilon_N / (8L)$, and the envelope sum gives $\|C_\times\|_\infty \leq C(m, t_0) \log_N$. Hence

$$\begin{aligned} \|(C_\times + R_N)^j - C_\times^j\|_\infty &\leq j \varrho_N (\|C_\times\|_\infty + \varrho_N)^{j-1}, \\ |\mathcal{P}_{N,j}| &\leq C(m, t_0, L, \Sigma_M(p), \bar{d})(1+t)^2 \varepsilon_N \log_N^{j-1}. \end{aligned} \tag{39}$$

This is the complete physical-edge contribution; it is not included in any Fourier alias term below.

We now decompose the surrogate expression exactly. Write

$$G_{N,j}(s, x) = \frac{1}{2L} \sum_{p \in B_N} b_N(s, p) e^{ipx}, \quad G_{\infty,j}(s, x) = \frac{1}{2L} \sum_{q \in \Gamma_\infty} b_\infty(s, q) e^{iqx},$$

and set $\tilde{b}(p_0, p) = \int_0^t b(s, p) e^{-ip_0 s} ds$. Parseval is

$$\|b\|_{2,\#}^2 := \frac{1}{2Lt} \sum_{p_0, p} |\tilde{b}(p_0, p)|^2 = \int_0^t \frac{1}{2L} \sum_p |b(s, p)|^2 ds. \tag{40}$$

Directly expanding the kernel powers and performing the time, lattice-space, and continuum-space integrals gives

$$Q_{N,j} = \frac{1}{2Lt} \sum_{p_0, p \in B_N} K_{N,j}^{\text{fold}}(p_0, p) |\tilde{b}_N(p_0, p)|^2, \quad (41)$$

$$Q_{\times,j} = \Re \left\{ \frac{1}{2Lt} \sum_{q_0, q \in \Gamma_\infty} K_{\times,j}(q_0, q) \overline{\tilde{b}_N(q_0, [q]_N)} \tilde{b}_\infty(q_0, q) \right\}, \quad (42)$$

$$Q_{\infty,j} = \frac{1}{2Lt} \sum_{q_0, q \in \Gamma_\infty} K_{\infty,j}(q_0, q) |\tilde{b}_\infty(q_0, q)|^2. \quad (43)$$

In (42) the summand is zero when $K_{\times,j} = 0$. The displayed real part is essential: the negative Nyquist edge belongs to D_N while the positive edge does not, so conjugate exact outputs need not both occur in the surrogate sum.

Extend b_N by zero outside B_N and put $\delta b = b_N - b_\infty$ on B_N , $\delta b = -b_\infty$ outside. Splitting $q = p + 2n\mathfrak{R}_N$ in (42) and expanding the square in $\langle \delta b, K_{\infty,j} \delta b \rangle$ gives the identity

$$\begin{aligned} Q_{N,j} - 2Q_{\times,j} + Q_{\infty,j} &= \langle \delta b, K_{\infty,j} \delta b \rangle + \langle b_N, (K_{N,j}^{\text{fold}} - K_{\infty,j}) b_N \rangle_{B_N} \\ &\quad - 2\Re \langle b_N, (K_{\times,j} - K_{\infty,j}) b_{\infty, B_N} \rangle_{B_N} - 2\Re A_{\times,j}, \end{aligned} \quad (44)$$

where every bracket includes the factor $(2Lt)^{-1}$ and

$$A_{\times,j} := \frac{1}{2Lt} \sum_{p_0, p \in B_N, n \neq 0} K_{\times,j}(p_0, p + 2n\mathfrak{R}_N) \overline{\tilde{b}_N(p_0, p)} \tilde{b}_\infty(p_0, p + 2n\mathfrak{R}_N). \quad (45)$$

Only finitely many n occur, because a j -fold output of D_N lies in the spatial interval $[-j\mathfrak{R}_N, j\mathfrak{R}_N]$. Equations (44)–(45) are an algebraic equality; there is no unnamed tail term.

By (29) and Cauchy–Schwarz, each Fourier coefficient is repeated in (45) at most $2j + 1$ times, so

$$|A_{\times,j}| \leq C_j B_j'' \varepsilon_N^2 \log_N^{j-1} \|b_N\|_{2,\#} \|b_\infty\|_{2,\#}. \quad (46)$$

Using (26)–(28) in (44) therefore yields

$$|Q_{N,j} - 2Q_{\times,j} + Q_{\infty,j}| \leq B_j \|\delta b\|_{2,\#}^2 + C \varepsilon_N^2 \log_N^2 (\|b_N\|_{2,\#} + \|b_\infty\|_{2,\#})^2. \quad (47)$$

We record the coefficient estimate rather than referring to a generic “same tail count”. For a spatial sequence set $\|a\|_{1,L} = (2L)^{-1} \sum_k |a_k|$, $\|a\|_{2,L}^2 = (2L)^{-1} \sum_k |a_k|^2$, and $(a *_L b)(p) = (2L)^{-1} \sum_k a_k b_{p-k}$. Young’s inequality reads $\|a *_L b\|_{2,L} \leq \|a\|_{2,L} \|b\|_{1,L}$. The coefficient of h^n is $a^{*L n}$, folded modulo $2\mathfrak{R}_N$ on the lattice and exact in the continuum, and

$$a_N^{*L n} - a_\infty^{*L n} = \sum_{\ell=0}^{n-1} a_N^{*L \ell} *_L (a_N - a_\infty) *_L a_\infty^{*L (n-1-\ell)}. \quad (48)$$

Lemma 6.3(i) gives uniform ℓ_L^1 bounds. Lemma 6.3(iii) controls the middle factor in ℓ_L^2 . A folded alias or an output outside B_N satisfies $|k_1 + \dots + k_n| \geq \mathfrak{R}_N$, hence some $|k_i| \geq \mathfrak{R}_N/n$. From the D4 envelope, for $1 \leq n \leq 4$,

$$\frac{1}{2L} \sum_{|k| \geq \mathfrak{R}_N/n} |a_k^{(\infty)}(s)|^2 + \frac{1}{2L} \sum_{\substack{k \in B_N \\ |k| \geq \mathfrak{R}_N/n}} |a_k^{(N)}(s)|^2 \leq C(\eta, n) \bar{P}_M^2 2^{-r(1+\eta)}. \quad (49)$$

Indeed $|a_k|^2 \leq C\varepsilon_M \bar{P}_M^2 (1 + \varepsilon_M |k|)^{-2-\eta}$ and the sum-integral comparison from the lower limit $\varepsilon_M \mathfrak{R}_N/n = \pi 2^r/n$ gives $\int_{\pi 2^r/n}^\infty (1+u)^{-2-\eta} du = (1 + \pi 2^r/n)^{-1-\eta}/(1+\eta)$. Applying Young to (48), and placing the forced factor of an alias in (49), proves for every $n \leq 4$

$$\|b_N^{(h_N^n)} - b_\infty^{(h_\infty^n)}\|_{2,\#}^2 \leq Ct 2^{-r\theta_H}, \quad \theta_H = (1+\eta)/2, \quad (50)$$

where the left side includes the common-band difference, all folded aliases, and all missing continuum outputs. The same calculation and $|d_N - d_\infty| \leq C\varepsilon_N^2$ give, for the exact coefficients $G_{a,j}$,

$$\|b_N\|_{2,\#}^2 + \|b_\infty\|_{2,\#}^2 \leq C(1+t), \quad \|\delta b\|_{2,\#}^2 \leq C(1+t)^3 [2^{-r\theta_H} + \varepsilon_N^4]. \quad (51)$$

Insert (51) into (47), add the physical-edge term (39) through (38), and use

$$\varepsilon_N \log_N^2 = \varepsilon_M 2^{-r} [1 + \log(1/\varepsilon_M) + r \log 2]^2 \leq C(\varepsilon_M, \eta) 2^{-r\theta_H}, \quad (52)$$

which holds because $1 - \theta_H > 0$. We obtain

$$\mathbb{E}[(T_{N,j} - T_{\infty,j})^2] \leq C(1+t)^4 [2^{-r\theta_H} + \varepsilon_N^2 \log_N^2], \quad j = 1, 2, 3. \quad (53)$$

For $j = 0$, $I_N(G_{N,0}) - I_\infty(G_{\infty,0})$ is the zero spacetime coefficient of the two- and four-fold convolutions in $G_{N,0}, G_{\infty,0}$. Equations (48)–(49), followed by Cauchy–Schwarz in spacetime and $|d_N - d_\infty| \leq C\varepsilon_N^2$, give

$$|I_N(G_{N,0}) - I_\infty(G_{\infty,0})| \leq C(1+t)^2 [2^{-r\theta_H/2} + \varepsilon_N^2]. \quad (54)$$

Finally sum the four terms in (37), take square roots in (53), and use $\theta_H/2 = (1+\eta)/4$, $\log_N \geq 1$, and $\varepsilon_N \leq 1$. Enlarging the fixed-parameter constant and the harmless time power to $(1+t)^{5/2}$ gives exactly the asserted estimate. $\square$

7 Uniform Nelson bounds (A5)

Lemma 7.1 (pointwise lower bound). *For every N and every realization,*

$$\mathcal{U}_N \geq -6\lambda(2Lt) c_{N,L}^2 \geq -c_1 t (1 + \log(1/\varepsilon_N))^2, \quad c_1 = c_1(\lambda, m, L), \quad (55)$$

and the same bound holds for the shifted interaction $\mathcal{U}_N(\cdot + h)$ for any deterministic real shift h — reality being essential, the identity $:u^4:_{\mathbf{c}} = (u^2 - 3c)^2 - 6c^2$ being one between real numbers. The shifts used below, $h^{(N)}$ and $h^{(\infty)}$ of Definition 6.2, are real by part (a) of the remark following that definition.

Proof. $\mathcal{U}_N = \lambda \varepsilon_N \sum_x \int : \eta_N^4 :_{\mathbf{c}_{N,L}}$ (Definition 2.2 = V1 form) and pointwise $:u^4:_{\mathbf{c}} = (u^2 - 3c)^2 - 6c^2 \geq -6c^2$; sum and integrate ($\varepsilon_N \cdot 2r_N \cdot t = 2Lt$). The Wick constant obeys $c_{N,L} \leq C(m, L)(1 + \log(1/\varepsilon_N))$ by the band sum $\sum_{\Gamma_N} \gamma_N^{-1} \leq C \log$. The bound is evaluated pointwise in the field argument, hence is invariant under deterministic real shifts (for complex u the identity gives no lower bound at all: at $u = ia$ with a real it reads $a^4 + 6ca^2 + 3c^2$). $\square$

Theorem 7.2 (A5: uniform Nelson bounds). *For every $p < \infty$ and $t \in [t_0, T]$:*

$$\sup_N \|e^{-\mathcal{U}_N}\|_{L^p(\mathbb{P})} < \infty, \quad (56)$$

with a bound depending on $(p, \lambda, m, L, t_0, T, \varepsilon_M)$ — here and only here p is the Lebesgue exponent, the coarse momentum field of the symbol $\xi = \varepsilon_M q + ip_M$ being written p_M ; the dependence on ε_M enters through the fourth entry of (57), which is imposed for every admissible b and hence also in the unshifted case, and it is harmless because M is fixed once and for all before $N \rightarrow \infty$ — ; the same holds, with $\mathcal{U}_N$ replaced by $\mathcal{U}_N(\cdot + h^{(N)})$, uniformly over coarse symbols with $\Sigma_M(p) \leq R$ and $\bar{P}_M \leq R'$ (constants then also depend on R, R').

Proof. Write $b_N^* := c_1 t (1 + \log(1/\varepsilon_N))^2$, so $\mathbb{P}[\mathcal{U}_N \leq -b] = 0$ for $b > b_N^*$ by Lemma 7.1. Fix $b \leq b_N^*$ with

$$L_b := (b/2c_1 t)^{1/2} \geq \max(2(1 + \log 2), 3, 4 \log 2/c_2, 1 + \log(1/\varepsilon_M)) \quad (57)$$

(the four entries are exactly what the three steps below require: $2(1 + \log 2)$ gives $\varepsilon_{N_b} \leq \frac{1}{2}$, the standing hypothesis of Theorem 4.3; 3 gives $\varepsilon_{N_b} \leq e^{-1/2}$, where $\varepsilon \mapsto \varepsilon(1 + \log(1/\varepsilon))^{3/2}$ is increasing; $4 \log 2/c_2$ gives $2q \geq 2$ in the hypercontractive step; and $1 + \log(1/\varepsilon_M)$ gives $N_b \geq M$, needed for $r_b := N_b - M$ and $h^{(N_b)}$ in the shifted case. Smaller b are absorbed into the constant at the end, at the cost of a factor e^{pb_1} in the layer cake, b_1 the value of b at equality above), and choose the comparison scale $N_b \in [M, N]$ as the largest dyadic scale with $1 + \log(1/\varepsilon_{N_b}) \leq L_b$. This exists and satisfies $1 + \log(1/\varepsilon_{N_b}) > L_b - \log 2 \geq L_b/2$: the value at M is admissible by (57), the value at N is inadmissible since $b \leq b_N^*$ forces $1 + \log(1/\varepsilon_N) \geq \sqrt{2} L_b > L_b$, and each dyadic refinement increases $1 + \log(1/\varepsilon)$ by exactly $\log 2$, so the largest admissible scale $N_b < N$ exists and the next scale's violation gives the lower bound. In particular $\varepsilon_{N_b} \leq e^{1-L_b/2}$. Then $\mathcal{U}_{N_b} \geq -c_1 t L_b^2 = -b/2$ pointwise (Lemma 7.1), so

$$\{\mathcal{U}_N \leq -b\} \subseteq \{\mathcal{U}_N - \mathcal{U}_{N_b} \leq -b/2\}.$$

By Theorem 4.3 and the triangle inequality through $\mathcal{U}$, and since $\varepsilon \mapsto \varepsilon(1 + \log(1/\varepsilon))^{3/2}$ is increasing on $(0, e^{-1/2})$,

$$\rho_b := \|\mathcal{U}_N - \mathcal{U}_{N_b}\|_{L^2} \leq C[\varepsilon_N \log_N^{3/2} + \varepsilon_{N_b} \log_{N_b}^{3/2}] \leq C_\rho e^{-c_2 L_b} L_b^{3/2}, \quad c_2 = \frac{1}{2}.$$

$\mathcal{U}_N - \mathcal{U}_{N_b}$ is a sum of Wick chaoses of order ≤ 4 , so Nelson's hypercontractive moment comparison applies in the sum-of-chaoses form of Part I, Appendix C (proof of the h -uniform stability theorem, **thm:stability**: write $Y = \Gamma(e^{-\tau})Z$, $Z = \sum_r e^{r\tau} Y_r$, use $\|\Gamma(e^{-\tau})\|_{2 \rightarrow q} \leq 1$ for $q = 1 + e^{2\tau}$ and chaos orthogonality; for order ≤ 4 this yields $\|Y\|_{L^{2q}} \leq (2q - 1)^2 \|Y\|_{L^2}$), and with Chebyshev in L^{2q} ,

$$\mathbb{P}[\mathcal{U}_N \leq -b] \leq \left(\frac{2}{b}\right)^{2q} (2q - 1)^{4q} \rho_b^{2q} \leq \left[\frac{2}{b} (2q)^2 \rho_b\right]^{2q}.$$

Choose $(2q)^2 = e^{c_2 L_b/2}$: the bracket is $\leq \frac{2C_\rho}{b} L_b^{3/2} e^{-c_2 L_b/2} \leq e^{-1}$ for $b \geq b_1$, $b_1 = b_1(\lambda, m, L, t_0, T, \varepsilon_M)$, whence

$$\mathbb{P}[\mathcal{U}_N \leq -b] \leq \exp(-e^{c_2 L_b/4}) = \exp\left(-\exp\left(\frac{c_2}{4} (b/2c_1 t)^{1/2}\right)\right),$$

uniformly in N (for $b > b_N^*$ the probability is zero, which is smaller). Since $e^{c\sqrt{b}}$ grows faster than any polynomial in b , the tail exponent eventually dominates $2pb$, so

$$\mathbb{E}[e^{-p\mathcal{U}_N}] = \int_{-\infty}^{\infty} p e^{pb} \mathbb{P}[\mathcal{U}_N \leq -b] db \leq e^{pb_1} + \int_{b_1}^{\infty} p e^{pb} e^{-\exp(\frac{c_2}{4} (b/2c_1 t)^{1/2})} db < \infty$$

with an N -independent bound (layer-cake identity as in Part I, App. C; the $b < b_1$ range contributes e^{pb_1} since $\mathbb{P} \leq 1$). For the shifted version: Lemma 7.1 holds verbatim under any deterministic shift, and $\mathcal{U}_N(\cdot + h^{(N)}) - \mathcal{U}_{N_b}(\cdot + h^{(N_b)}) = (\mathcal{U}_N - \mathcal{U}_{N_b}) + (\Delta\mathcal{U}_N - \Delta\mathcal{U}_{N_b})$, where the second bracket is controlled by Proposition 6.5 at the two scales: with $r_b := N_b - M$, $2^{-r_b(1+\eta)/4} = (\varepsilon_{N_b}/\varepsilon_M)^{(1+\eta)/4} \leq C(\varepsilon_M) e^{-L_b(1+\eta)/8}$, so the total obeys the same bound with $c_2 = (1+\eta)/8$ and a constant uniform over $\Sigma_M(p) \leq R$, $\bar{P}_M \leq R'$ (Lemma 6.3); the argument then runs verbatim with this smaller c_2 . $\square$

8 Acceptance and claim ledger

V0 acceptance: Theorem 5.1 proves the constant identified numerically ($0.0384461803\dots$ to 3.6×10^{-10}); no other V0 item was scheduled against V2. The A4/A5 outputs consumed by A6 are: Theorem 4.3, Proposition 6.5, Theorem 7.2, and Lemma 2.1/6.3; the convention lock for the gate's right-hand side is Corollary 5.2.

claim	status	ground
Same-noise coupling realizes the V1 lattice ensemble	proved	Prop. 1.2 (Mittag–Leffler sum (6))
Recombination; $d_N(t)$ finite, $\rightarrow d_\infty(t)$, $O(\varepsilon_N^2)$	proved	Lemma 2.1
Existence of the continuum Wick integrals Y_2, Y_4	proved	Prop. 2.4 (monotone mode sums)
Envelope $\hat{c} \leq \frac{\pi^{2/4}}{ \kappa ^2 + m^2}$; convolution decay; summability	proved	Lemma 3.1
Alias sums $\leq C(m, t_0)tL\varepsilon_N^2 \log^3 N$	proved	Lemma 4.2
Symmetric-edge correction: one-, two-, and four-edge sectors retained; exactly three impossible; $\Delta_{4e} \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5$	proved	Proposition 1.2, equations (3)–(5)
$\ \mathcal{U}_N - \mathcal{U}\ _{L^2} \leq C(1+t)(1+L)\varepsilon_N \log^{3/2} N$	proved	Theorem 4.3 (second chaos: exact-square structure)
Dictionary $\delta_\infty = \frac{1}{2\pi} \log \frac{4}{\pi}$ with rate $O(\varepsilon_N)$, explicit constant	proved	Theorem 5.1; V0 confirmation
Convention lock: gate polynomial is plain φ^4 (no counterterm; the two tadpole offsets cancel)	proved	Corollary 5.2, Remark 5.3
Closed form $H(s) = \frac{\rho}{2} \frac{\sinh(\omega(s-t/2))}{\sinh(\omega t/2)}$; $ H \leq \rho /2$; ω -derivative bound	proved	Lemma 6.1
Uniform coarse-data shift bounds ($\Sigma_M(p)$); symbol convergence at rate $2^{-r(1+\eta)/2}$	proved	Lemma 6.3
Folded lattice, exact-output mixed, and continuum power symbols: uniform bounds, fundamental-band differences, and high-output tails	proved	Lemma 6.4, equations (27)–(29)
Shifted-interaction L^2 comparison, $O(2^{-r(1+\eta)/4} + \varepsilon_N \log N)$; exact four-term quadratic partition for the real asymmetric-band surrogate, with explicit $A_{\times,j}$	proved	Prop. 6.5, equations (44)–(45)
Uniform Nelson bounds, plain and shifted	proved	Theorem 7.2

External imports used (verbatim, with source locations):

- free-rate document (file `lamport_part_ii_free_fixed_coarse_rate`): Lemma 2.1, eq. (2.4) ($D4$ envelope $|P_r(\ell)|^2 \leq C_D(1 + |\ell|)^{-2-\eta}$, $\eta = 2 - \log_2 3$, proved for $|\ell| \leq \pi 2^r$ and, for P_∞ , on all of $\mathbb{R}$); Lemma 3.1, eq. (3.1) ($0 \leq \gamma - \gamma_N \leq \varepsilon_N^2 k^4 / (24m)$, under $|\varepsilon_N k| \leq \pi$); and the display (3.5) *inside the proof* of that lemma (its step LS 1.1), $0 \leq k^2 - 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2) \leq \varepsilon_N^2 k^4 / 12$, likewise valid under $|\varepsilon_N k| \leq \pi$ — which holds throughout since $k \in \Gamma_N$ — consumed in Lemma 3.2;
- model sheet (file `lamport_part_ii_model_sheet`): the conventions of §1–§2 and, in Lemma 6.3, the mask coefficients (2.1), from which $L_{m_0} \leq \frac{1}{\sqrt{2}} \sum_n n |h_n| = (3 + \sqrt{3})/4$;
- V1 (file `v1_thermal_representations`): Lemma 5.2 (`lem:interp`, interpolant characterization and uniqueness), Theorem 5.4 (`thm:meanshift`, Wick-binomial shifted form), Remark 5.5 (`rem:ensemble`, the periodic ensemble matched in Proposition 1.2);
- Part I (file `part_i/main.tex`), Appendix C, proof of the h -uniform stability theorem (labels `app:huniform`, `thm:stability`): the $\Gamma(e^{-\tau})$ hypercontractive moment comparison for sums of Wick chaoses, and the layer-cake assembly of $\mathbb{E}[e^{-p\mathcal{U}}]$ from the doubly-exponential tail.

The $D4$ mask Lipschitz constant $L_{m_0} \leq (3 + \sqrt{3})/4$ is derived inline in Lemma 6.3 from the model-sheet mask coefficients. No other external facts are consumed.

Phase V2 gate: A4(a), A4(b), A4(c), and A5 are closed; constants explicit; the δ_∞ value matches V0. Remaining phases: V3 (Statement A1 with the (A1-iv) FK clause) and V4 (A6–A8 assembly and the adversarial pass).